

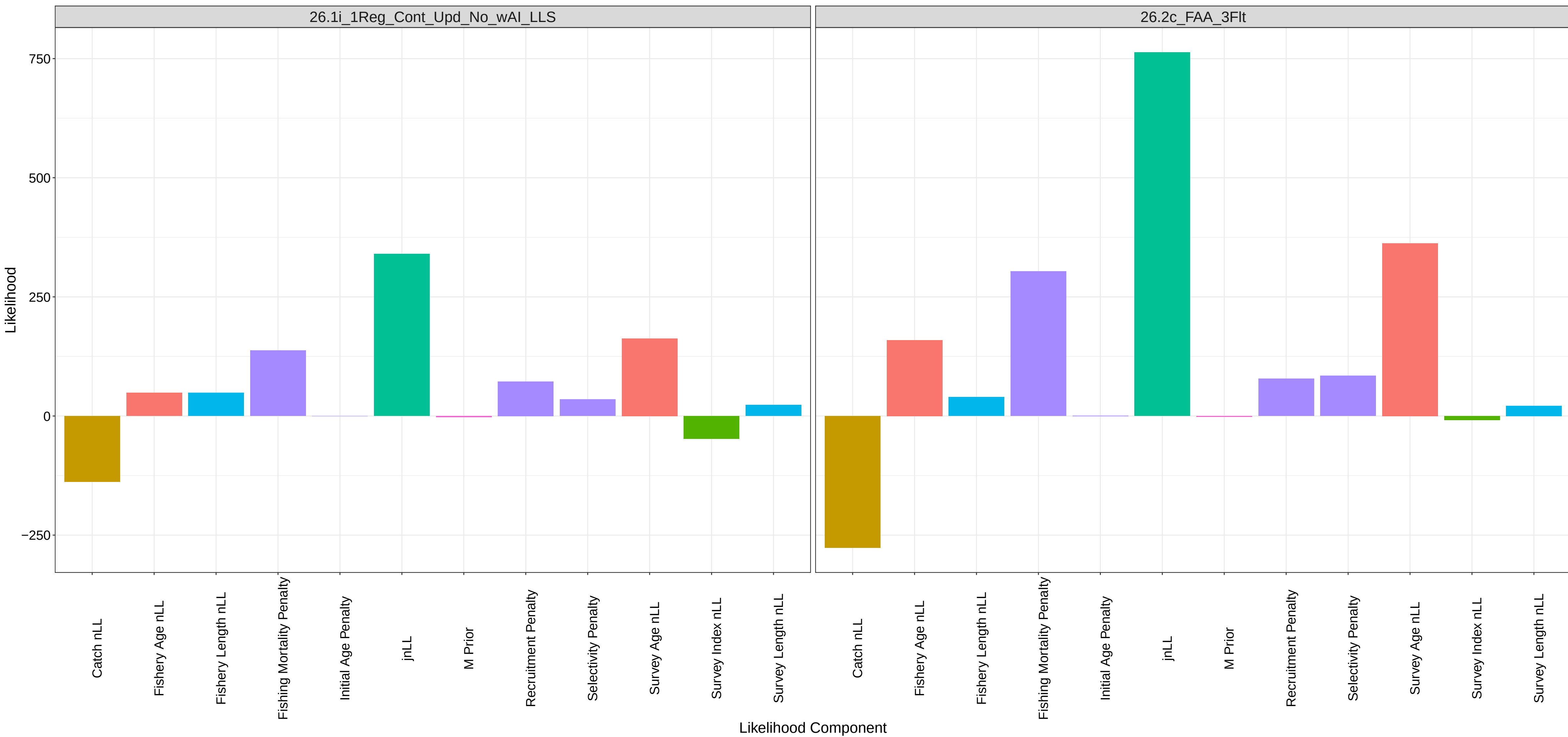
|   | Model                          | Region | Terminal_SSB | Terminal_SSB0 | Terminal_F | Catch_Advice | B_Ref_Pt | F_Ref_Pt | B_over_B_Ref | B_over_DynB_Ref | F_over_F_Ref |
|---|--------------------------------|--------|--------------|---------------|------------|--------------|----------|----------|--------------|-----------------|--------------|
| 1 | 26.1i_1Reg_Cont_Upd_No_wAI_LLS | 1      | 203.6004     | 357.3176      | 0.04932641 | 37.93084     | 133.9521 | 0.07808  | 1.51995      | 0.56980         | 0.63177      |
| 2 | 26.2c_FAA_3Flt                 | 1      | 203.8097     | 358.5356      | 0.05532669 | 34.44823     | 119.0156 | 0.08357  | 1.71246      | 0.56845         | 0.66204      |

| Model |                                | jnLL     | M Prior   | Recruitment Penalty | Selectivity Penalty | Catch nLL | Fishing Mortality Penalty | Survey Index nLL | Initial Age Penalty | Survey Age nLL | Fishery Age nLL | Survey Length nLL | Fishery Length nLL |
|-------|--------------------------------|----------|-----------|---------------------|---------------------|-----------|---------------------------|------------------|---------------------|----------------|-----------------|-------------------|--------------------|
| 1     | 26.1i_1Reg_Cont_Upd_No_wAI_LLS | 340.1992 | -2.406027 | 72.37126            | 34.93611            | -138.3564 | 137.4845                  | -47.688764       | 0.1682512           | 162.8081       | 48.83554        | 23.32842          | 48.71815           |
| 2     | 26.2c_FAA_3Flt                 | 763.2411 | -1.802370 | 78.49590            | 84.53175            | -276.6025 | 303.7534                  | -8.836993        | 0.7676920           | 362.5001       | 159.24061       | 21.46507          | 39.72842           |

# Negative Log-Likelihood

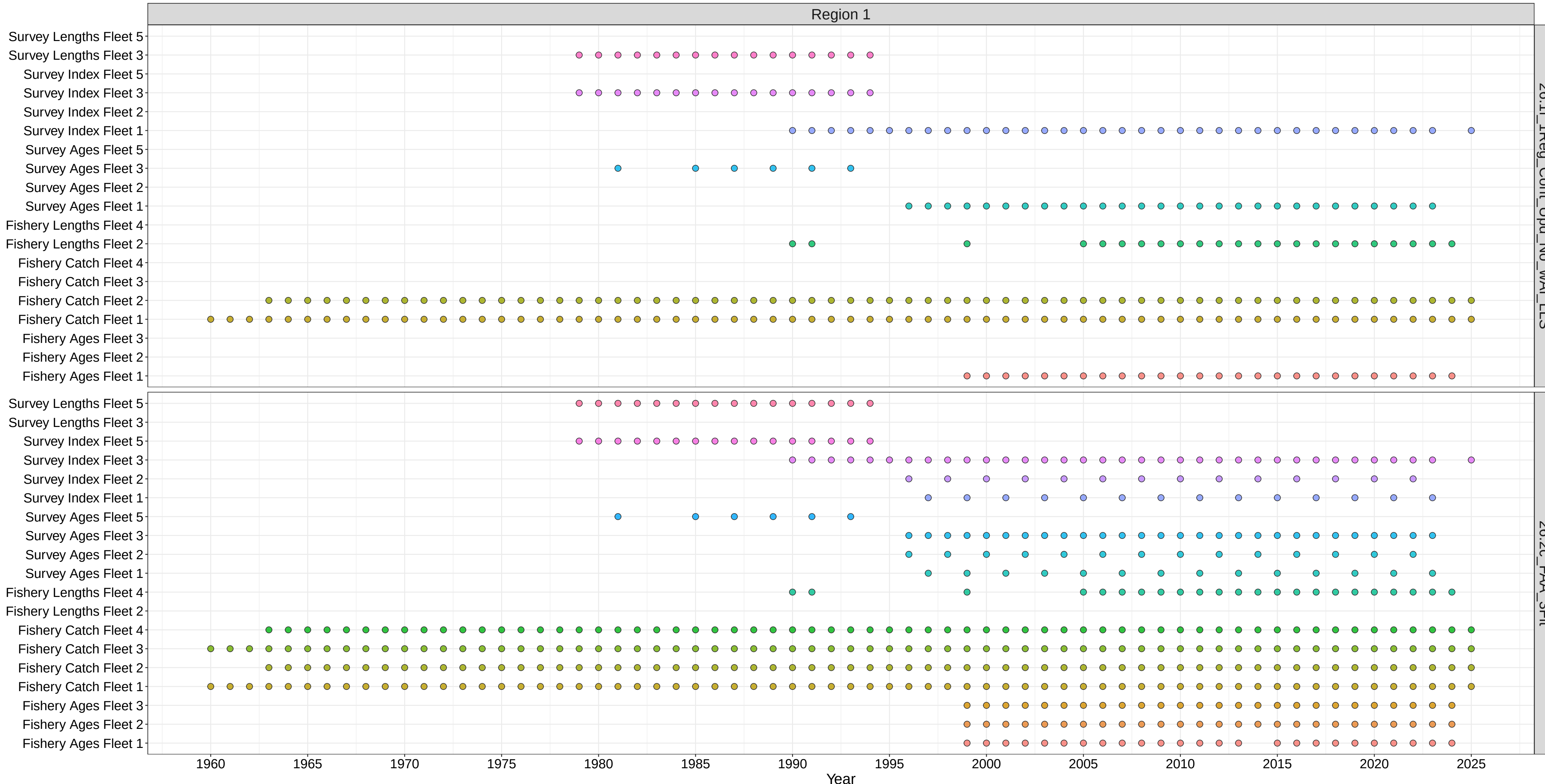
Type

- Age
- Catch
- Index
- jnLL
- Length
- Prior
- Penalty



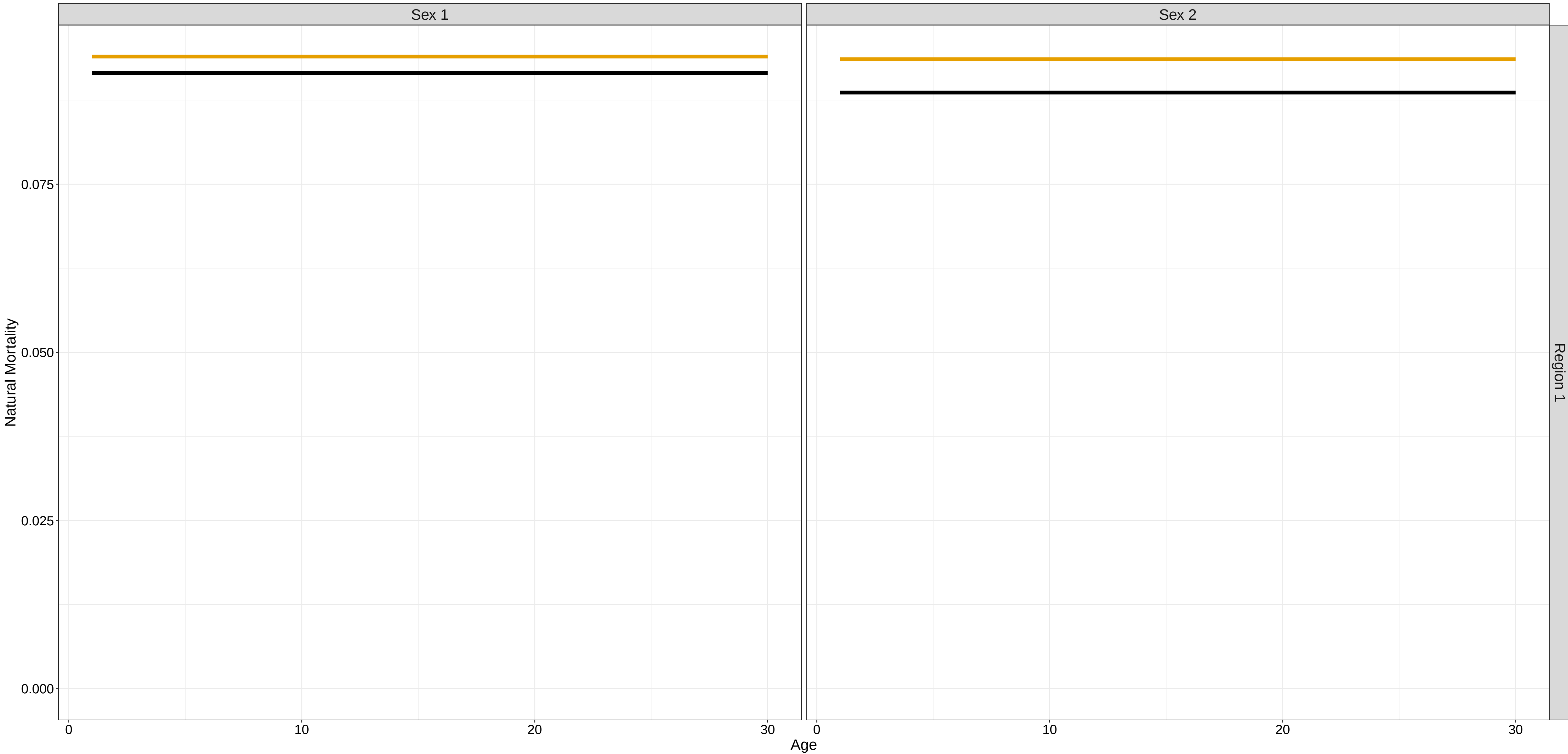


Data



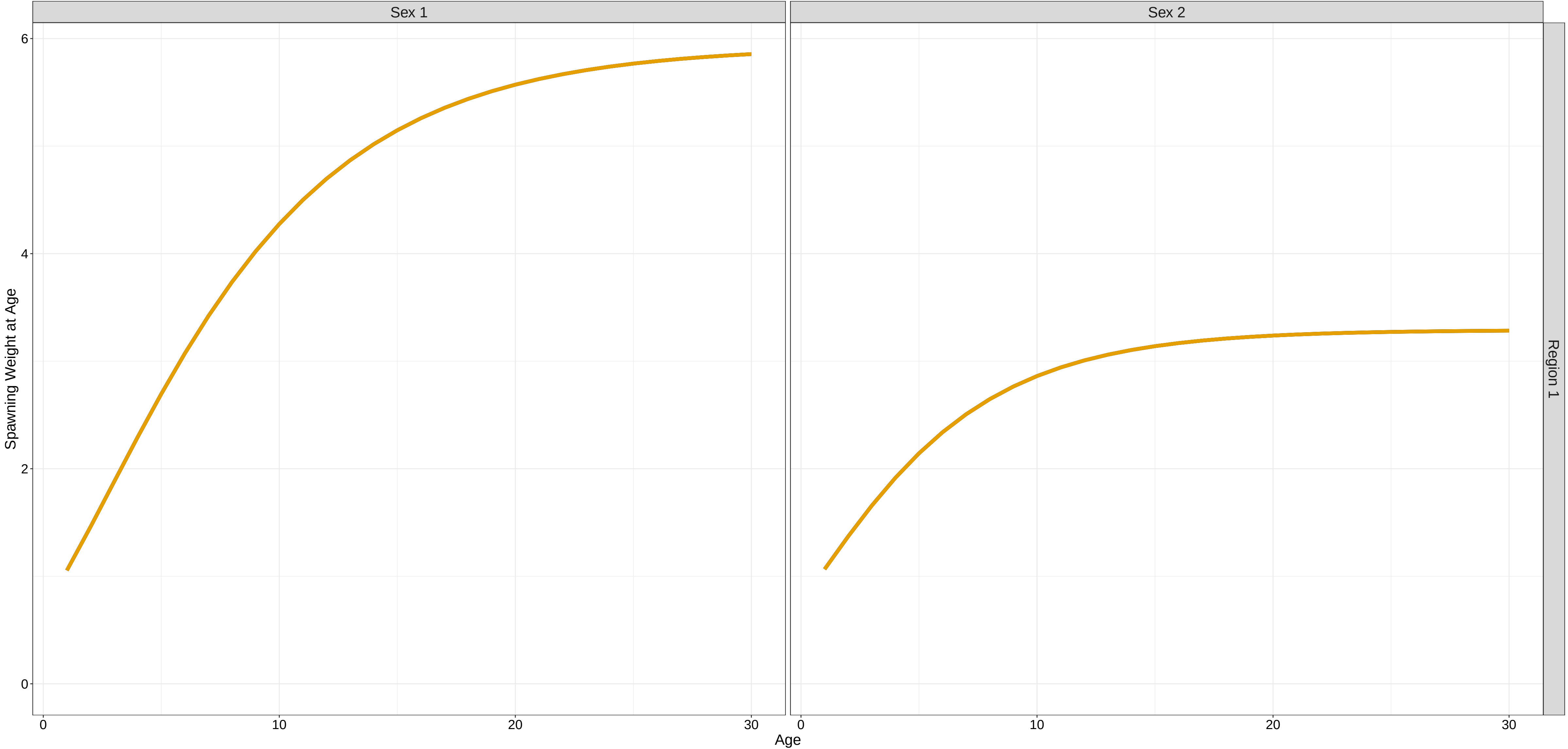
# Natural Mortality

Model — 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS — 26.2c\_FAA\_3Flt



Weight-at-Age

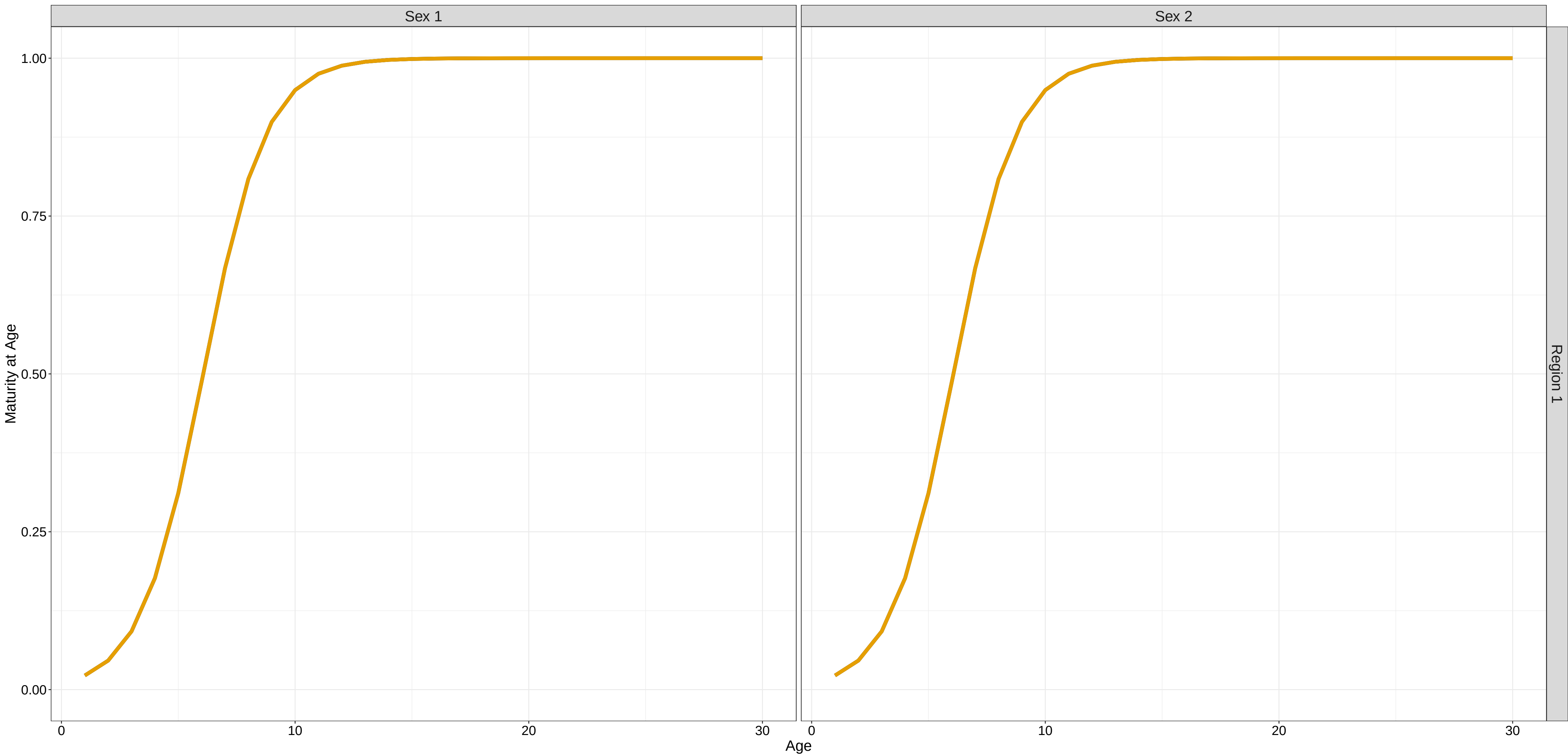
Model 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS 26.2c\_FAA\_3Flt





# Maturity

Model 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS 26.2c\_FAA\_3Flt



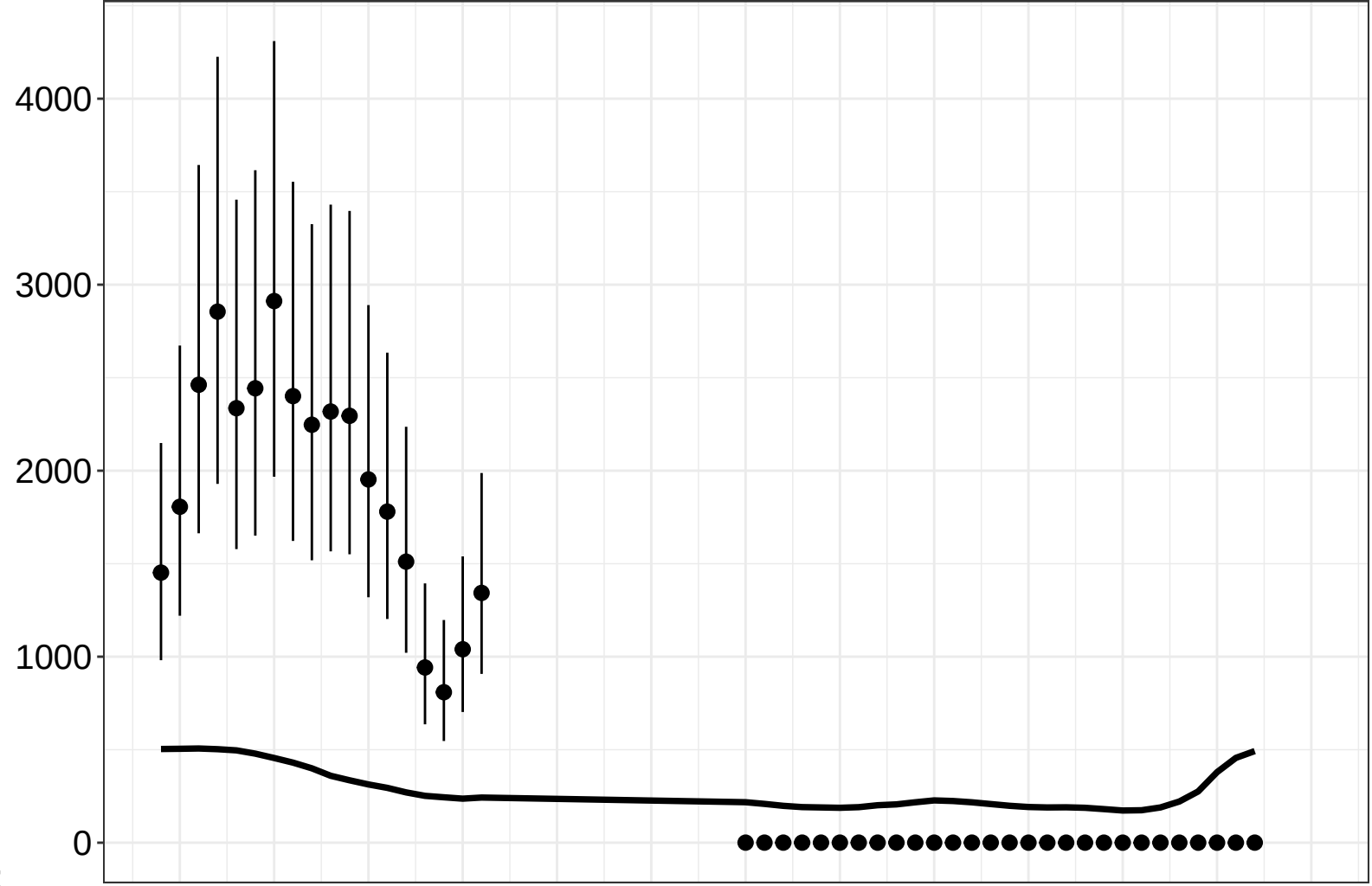
Region 1

# Index Fits

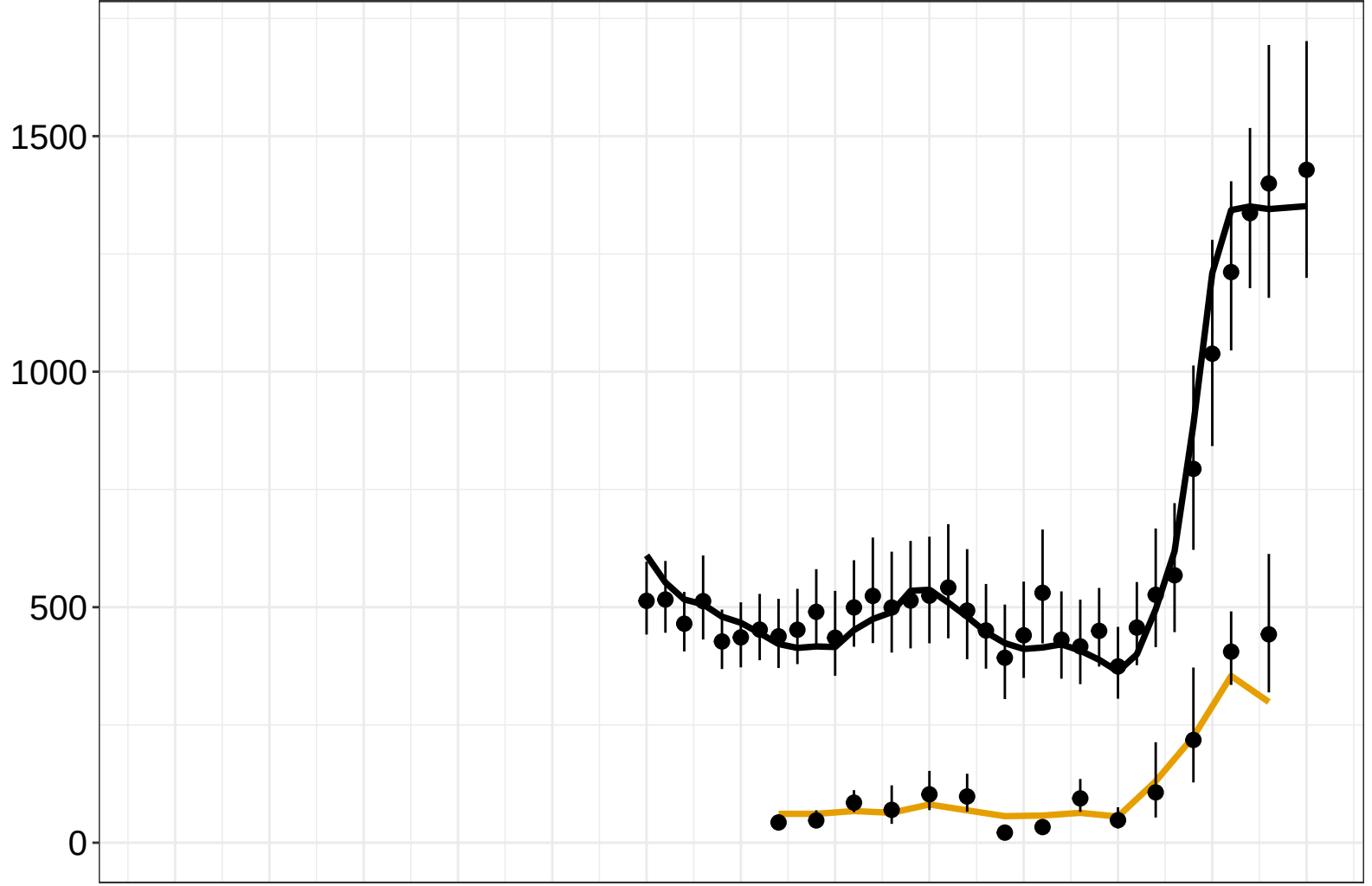
Model — 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS — 26.2c\_FAA\_3Flt

Index

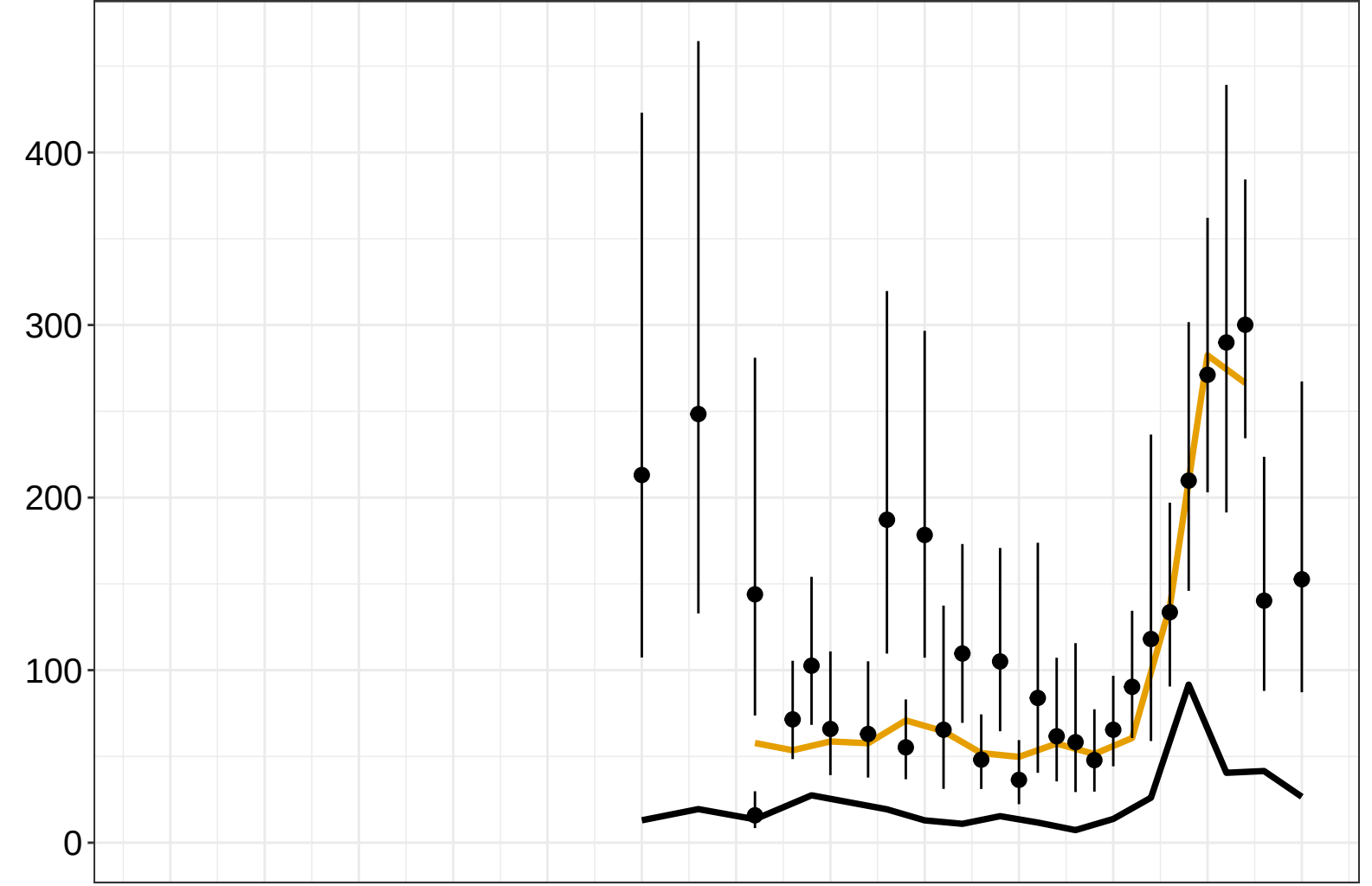
JPN Fishery CPUE



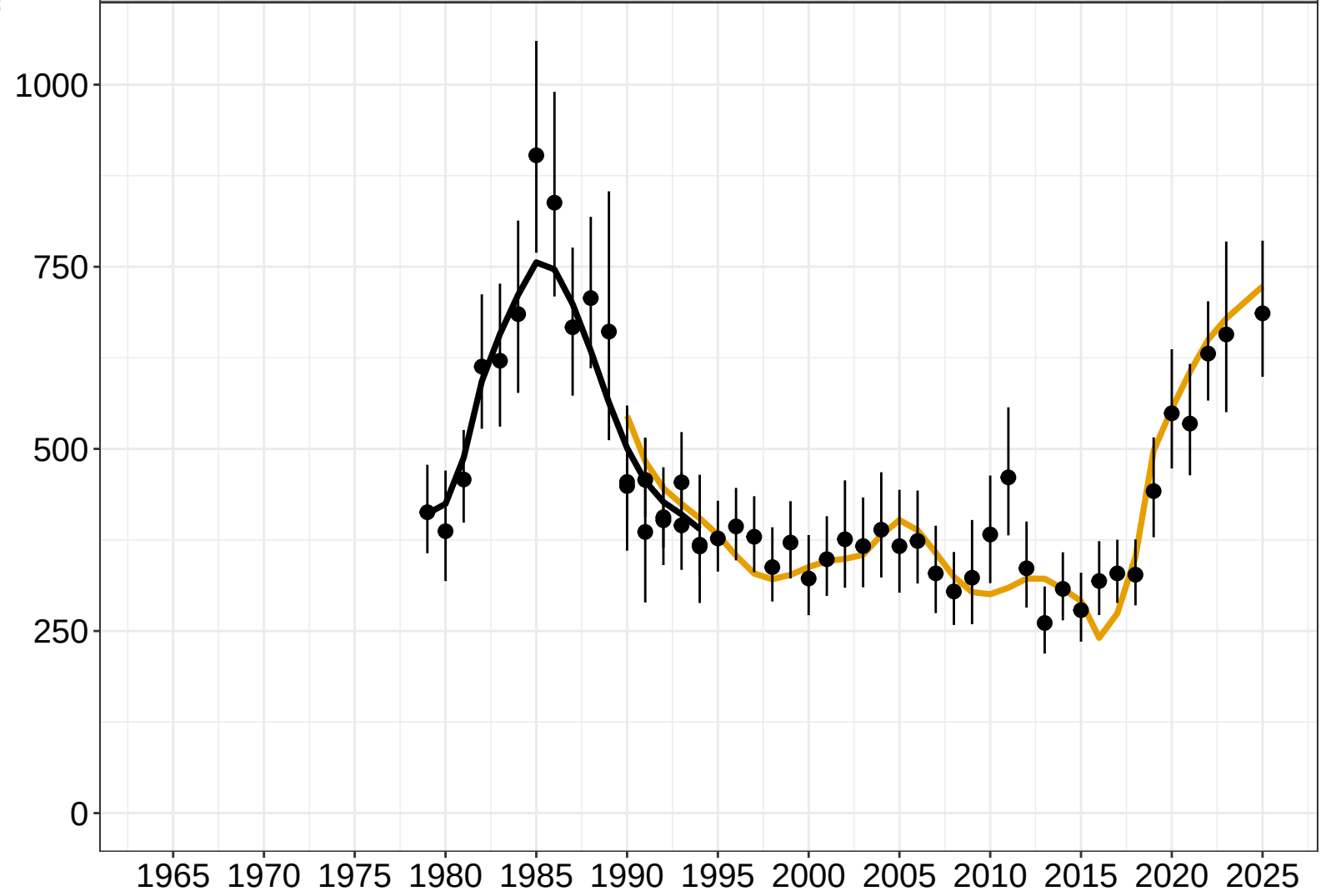
BS LL Survey RPNs



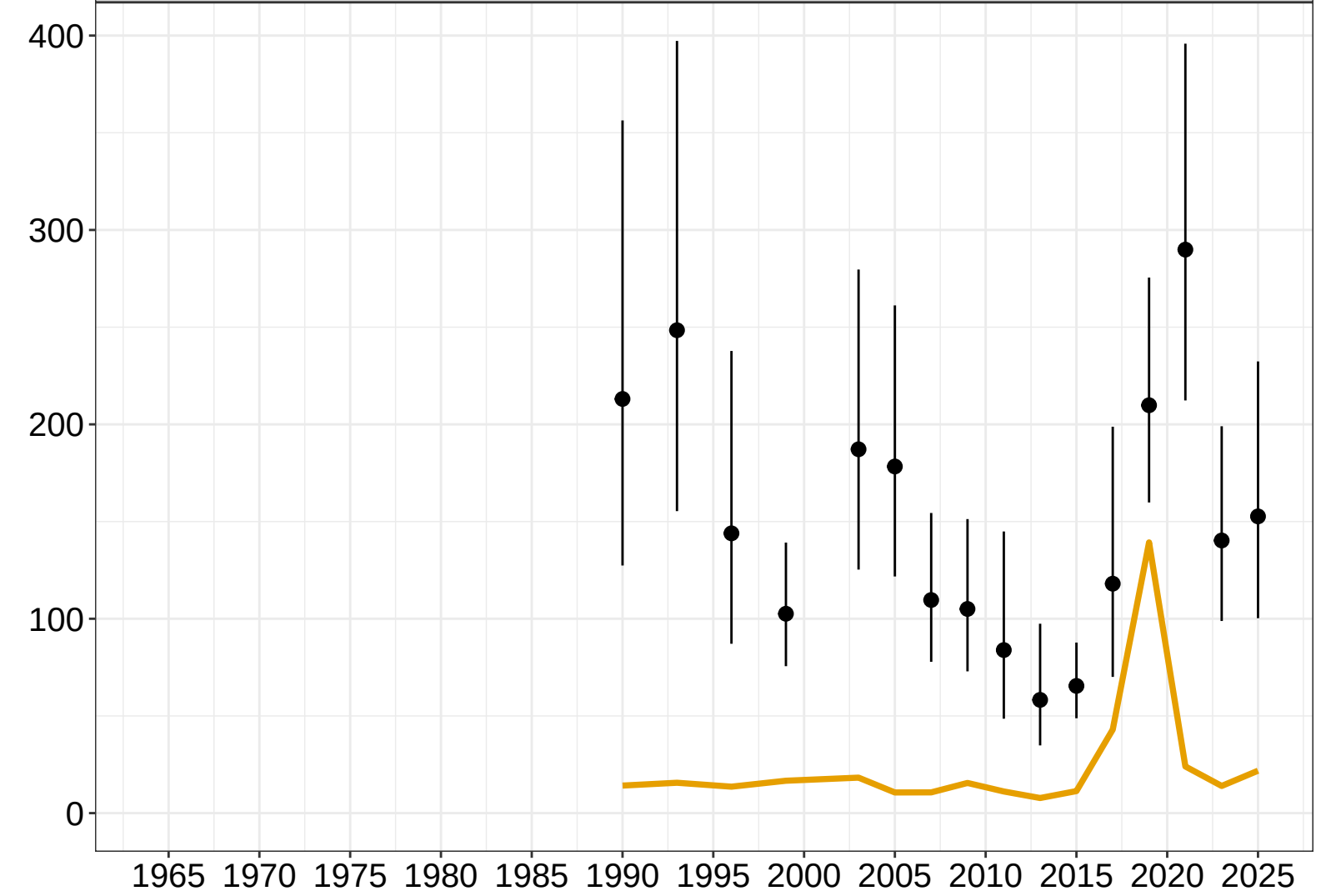
AI LL Survey RPNs



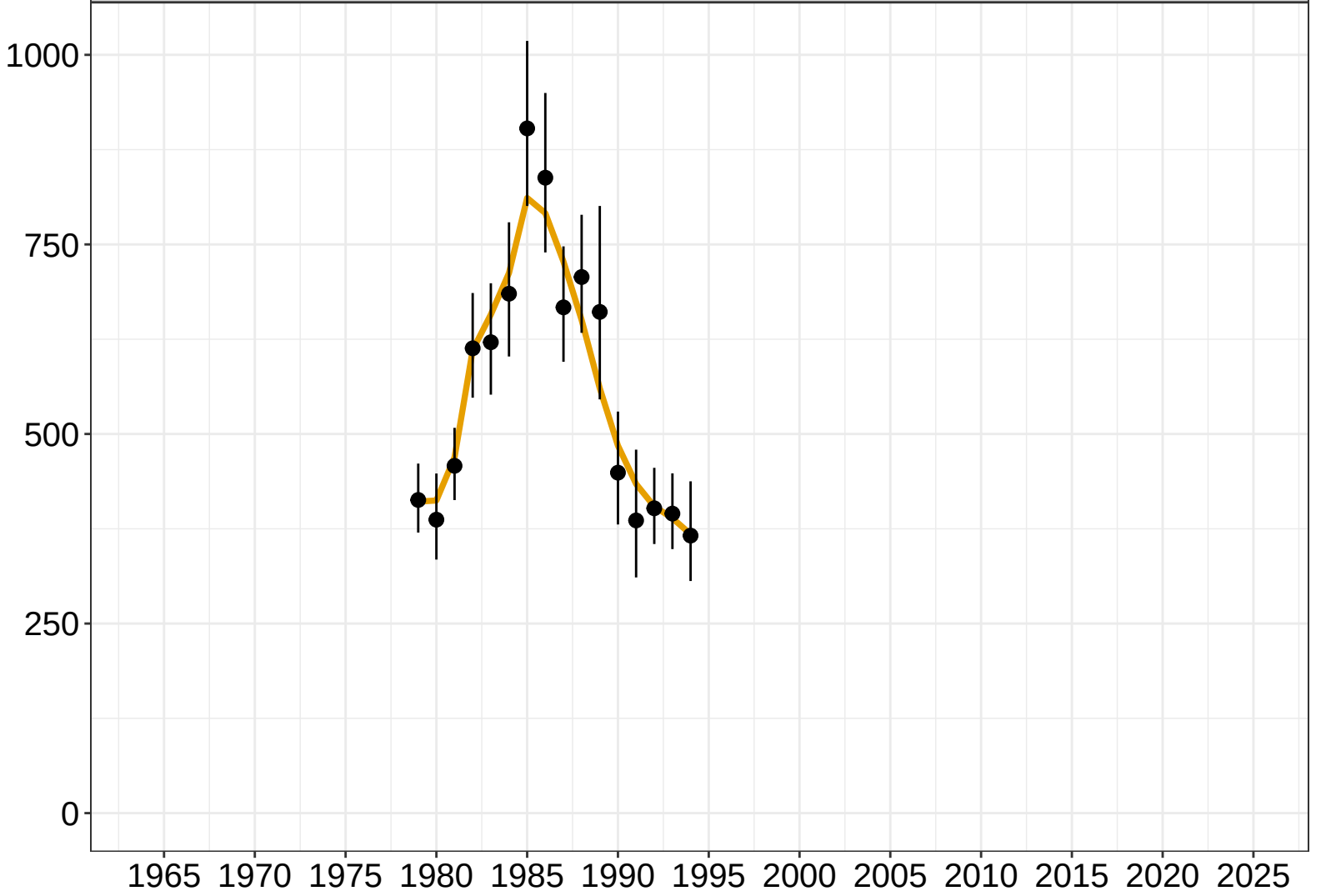
GOA LL Survey RPNs



GOA Trawl Survey (kt)



JPN LL Survey RPNs

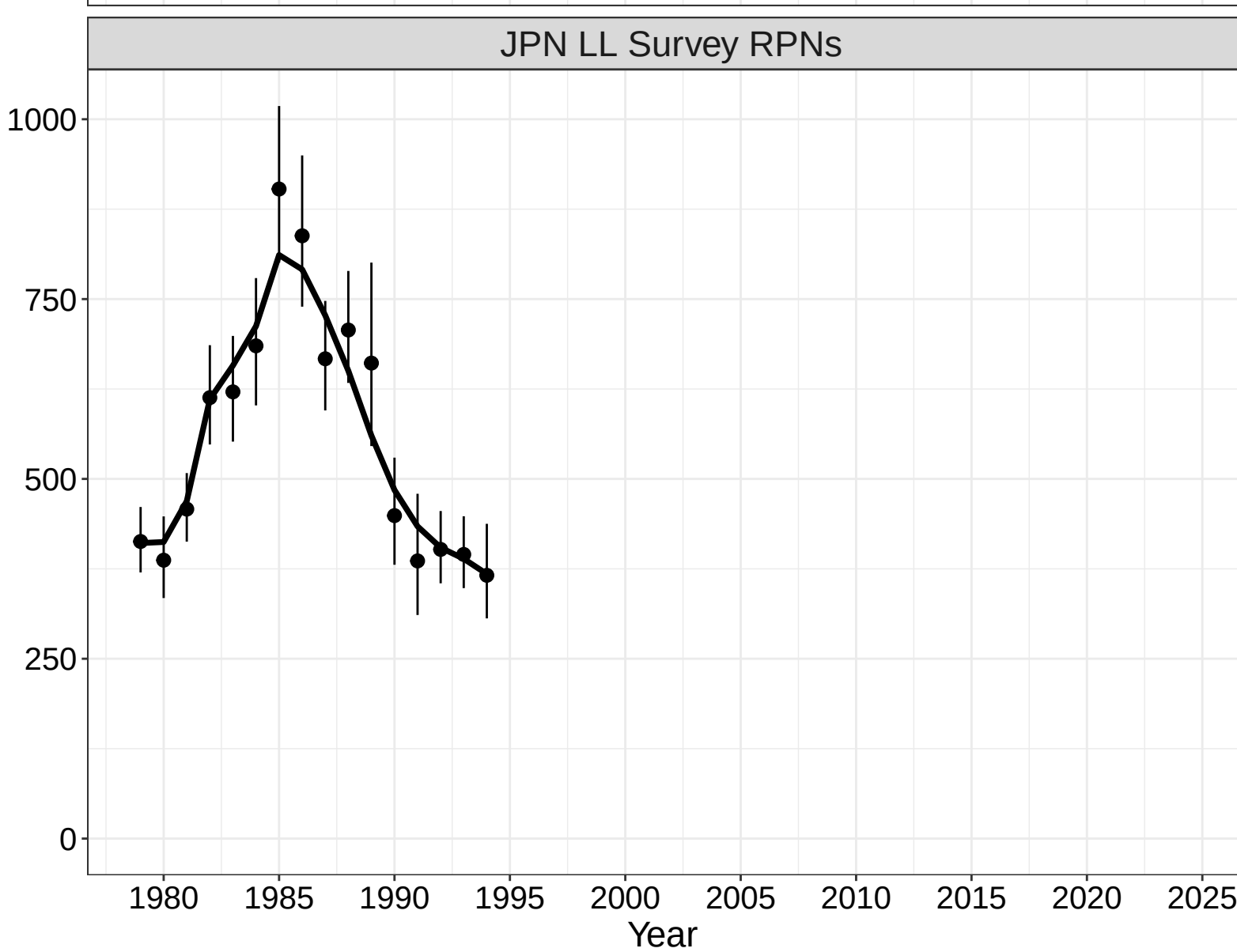
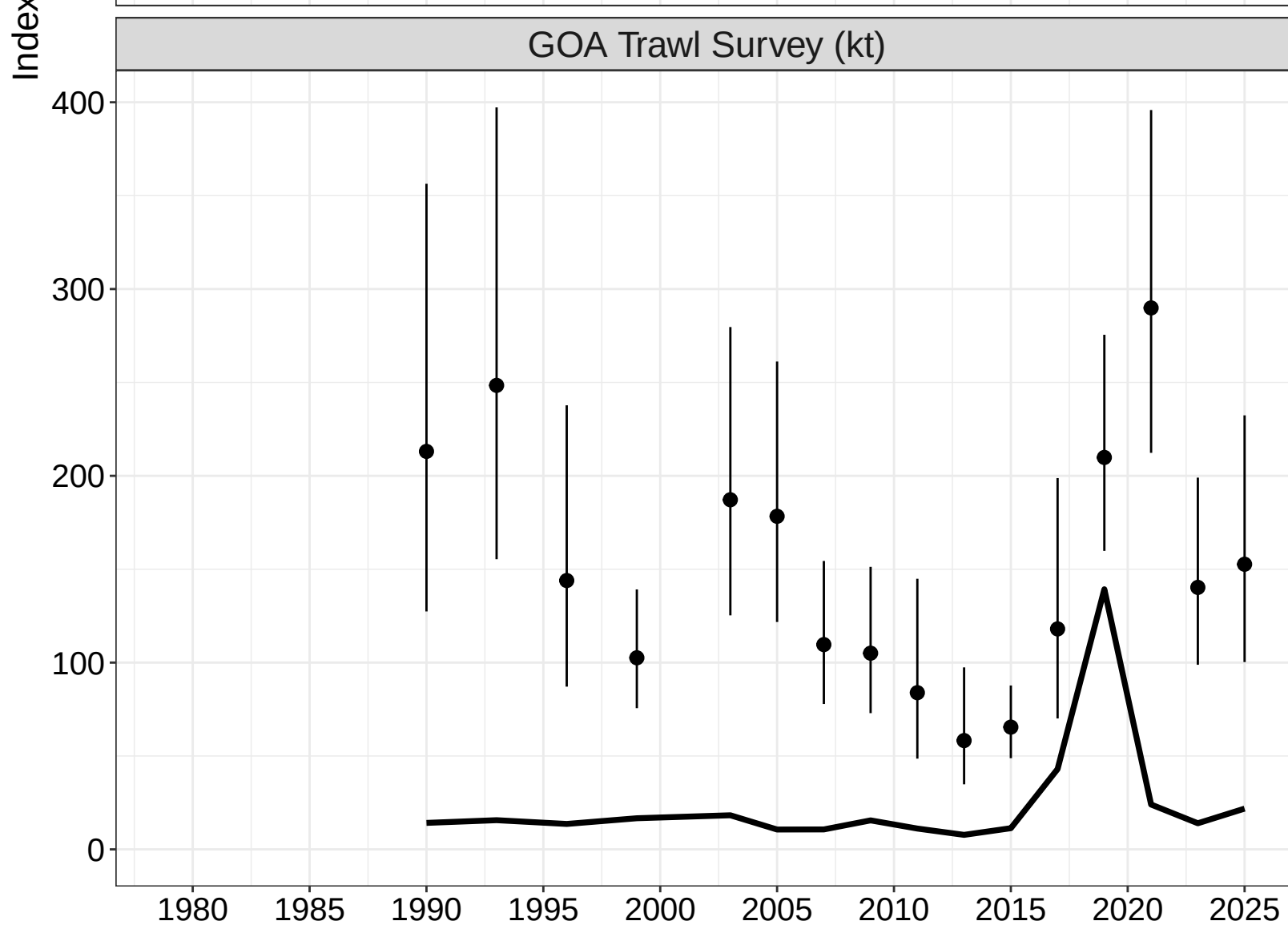
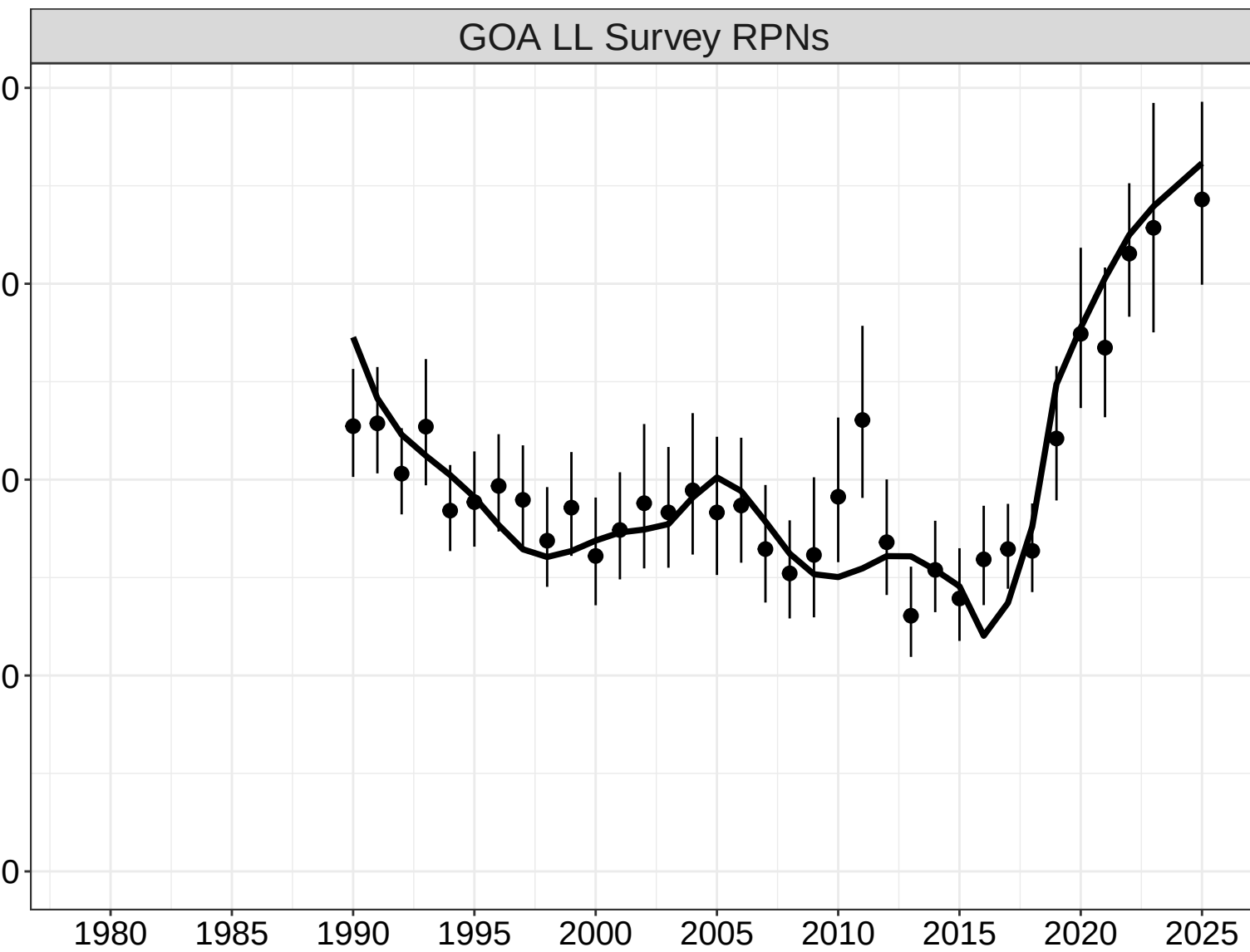
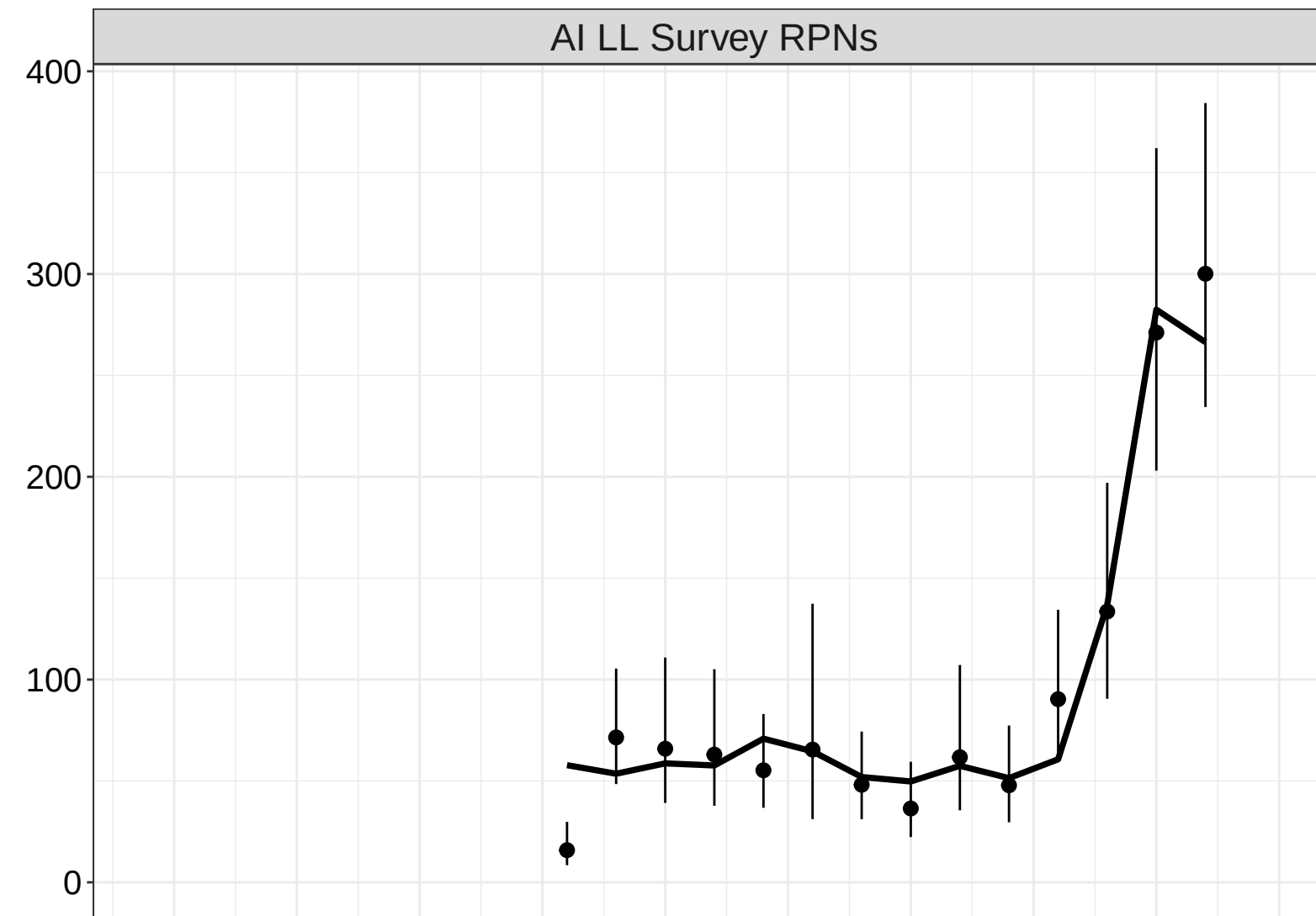
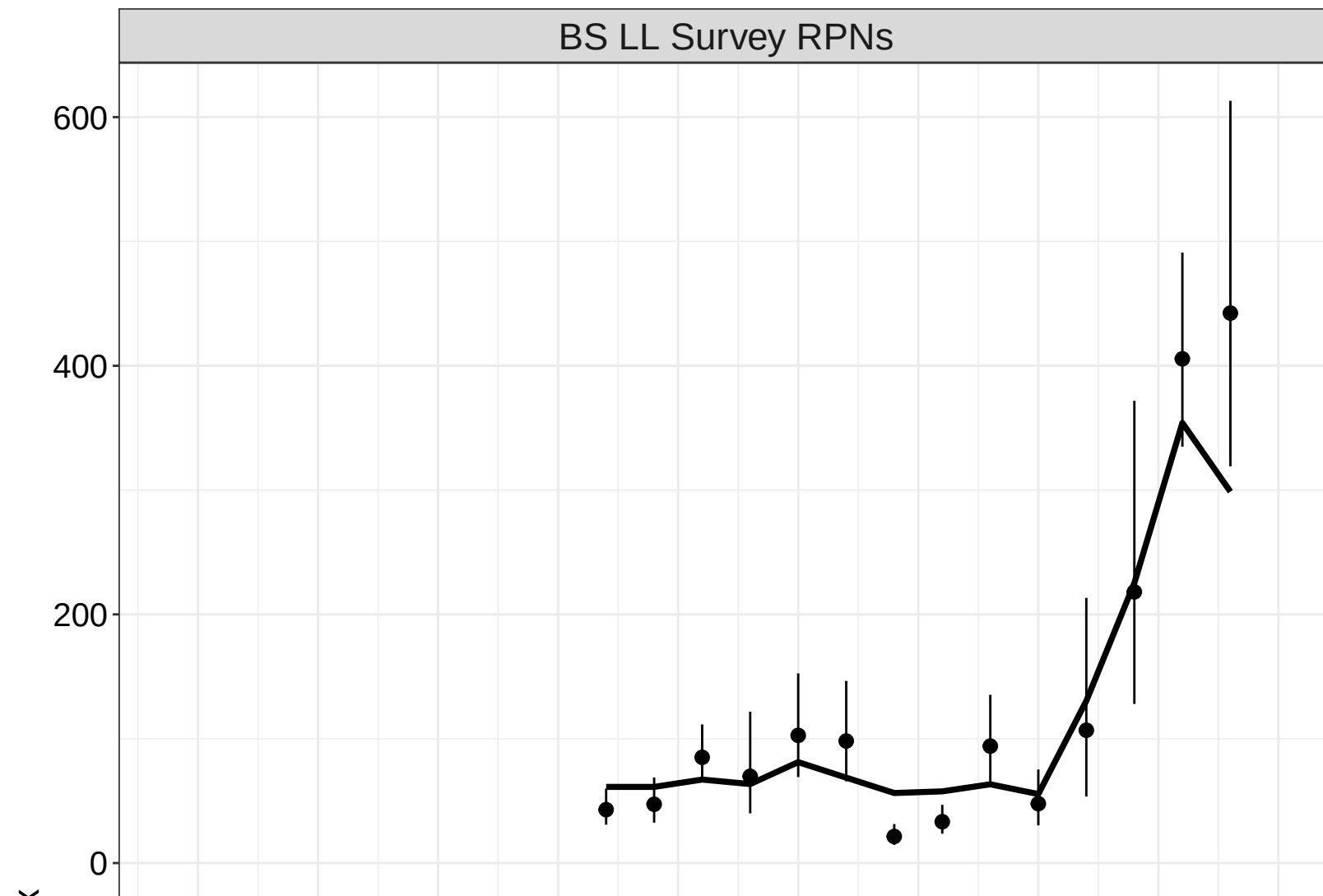


Year



# Index Fits

Model — 26.2c\_FAA\_3Flt



Catch

Model 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS 26.2c\_FAA\_3Flt

Region 1

Fleet 1

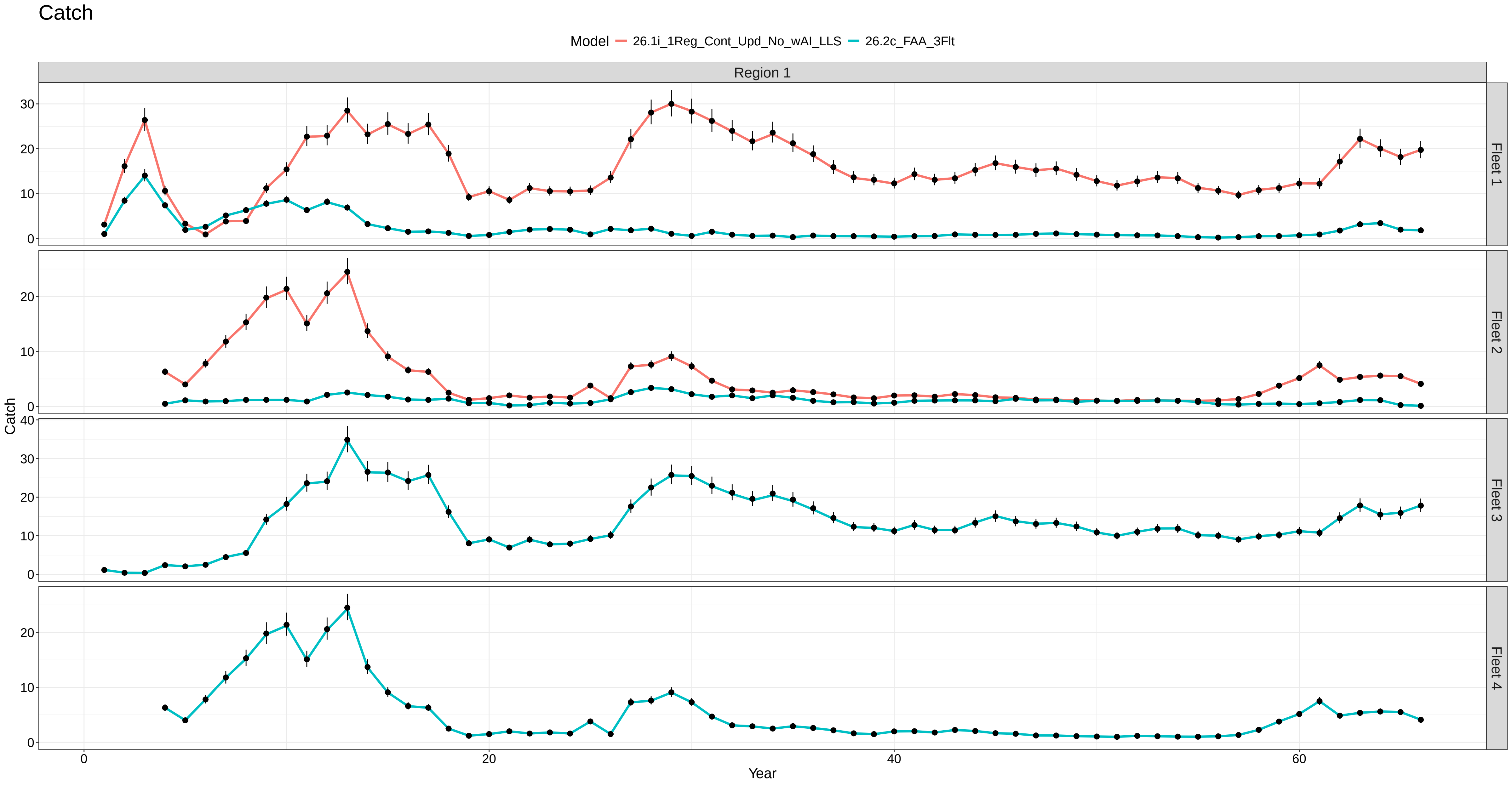
Fleet 2

Fleet 3

Fleet 4

Catch

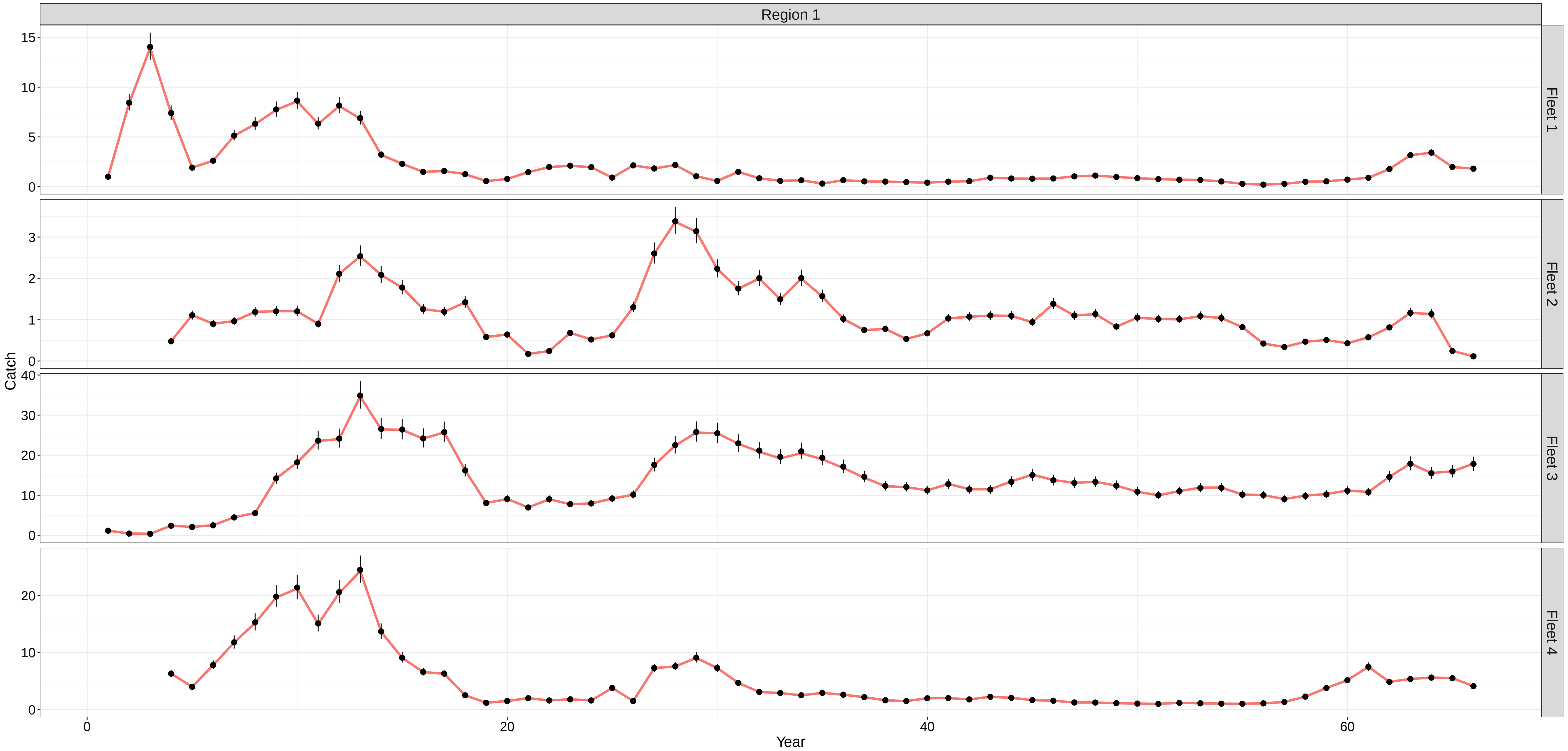
Year



Catch

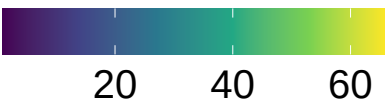
Model 26.2c\_FAA\_3Flt

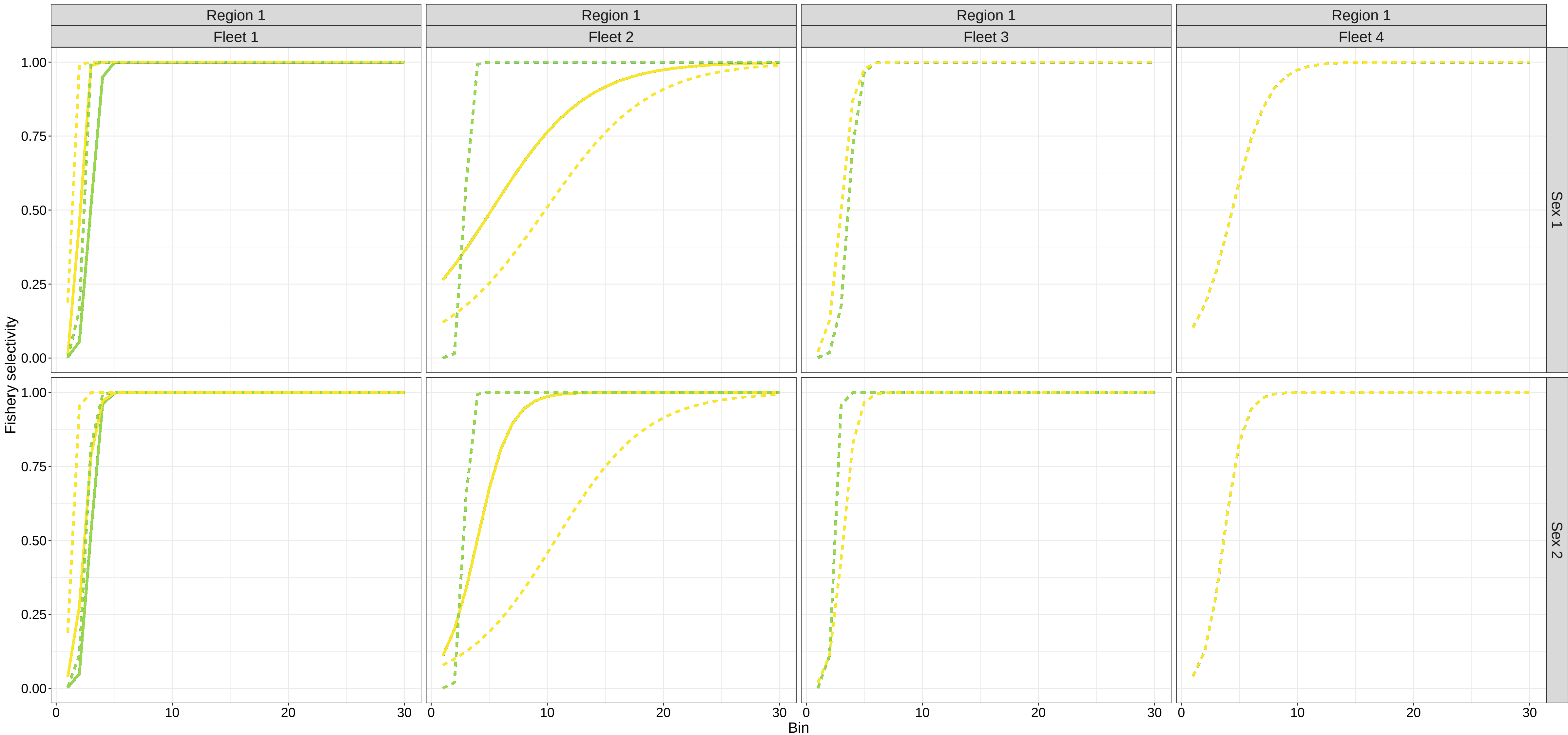
Region 1





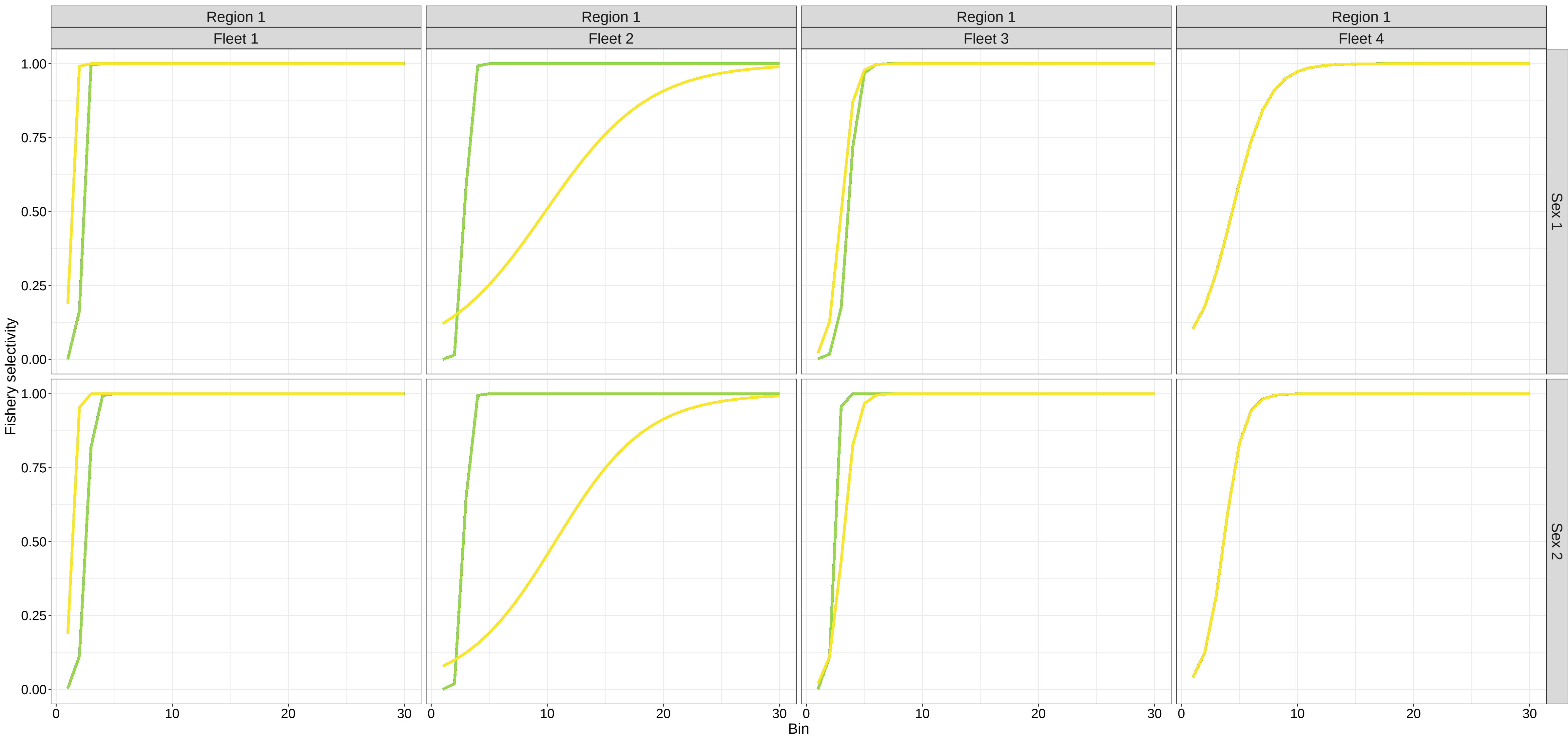
# Fishery Selectivity

Model — 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS - - - 26.2c\_FAA\_3Flt Year 



# Fishery Selectivity

Model — 26.2c\_FAA\_3Flt

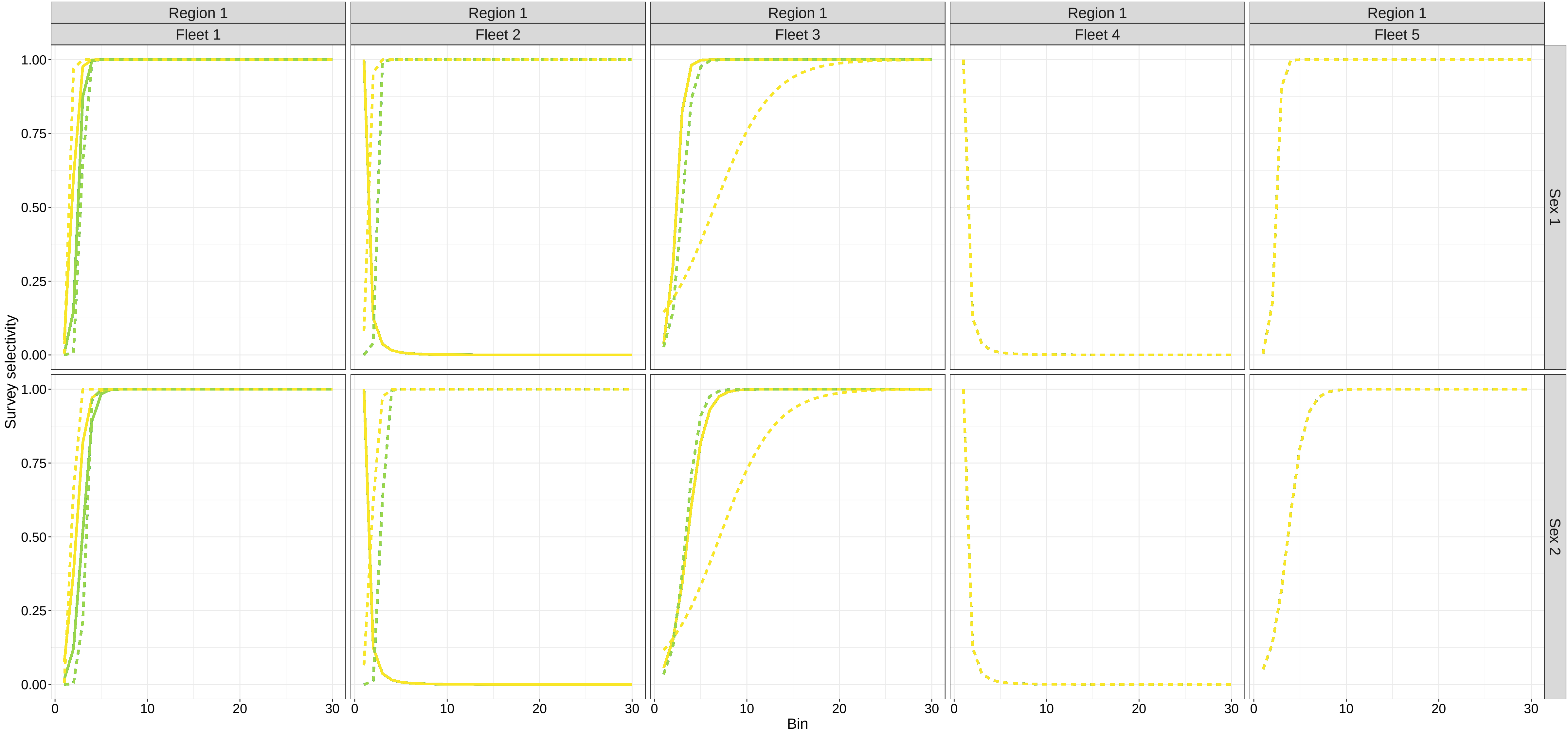


# Survey Selectivity

Model

— 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS    - - - 26.2c\_FAA\_3Flt

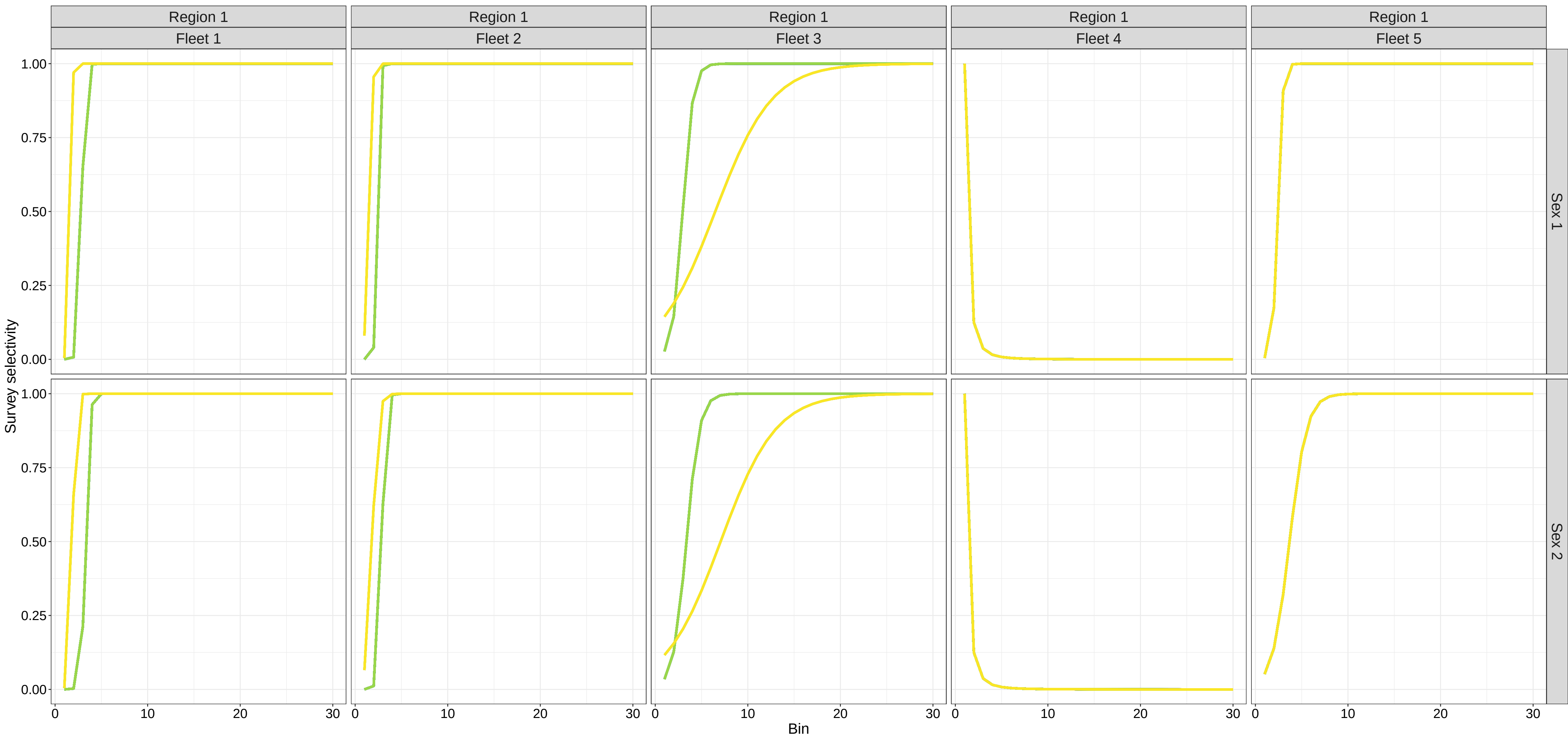
Year





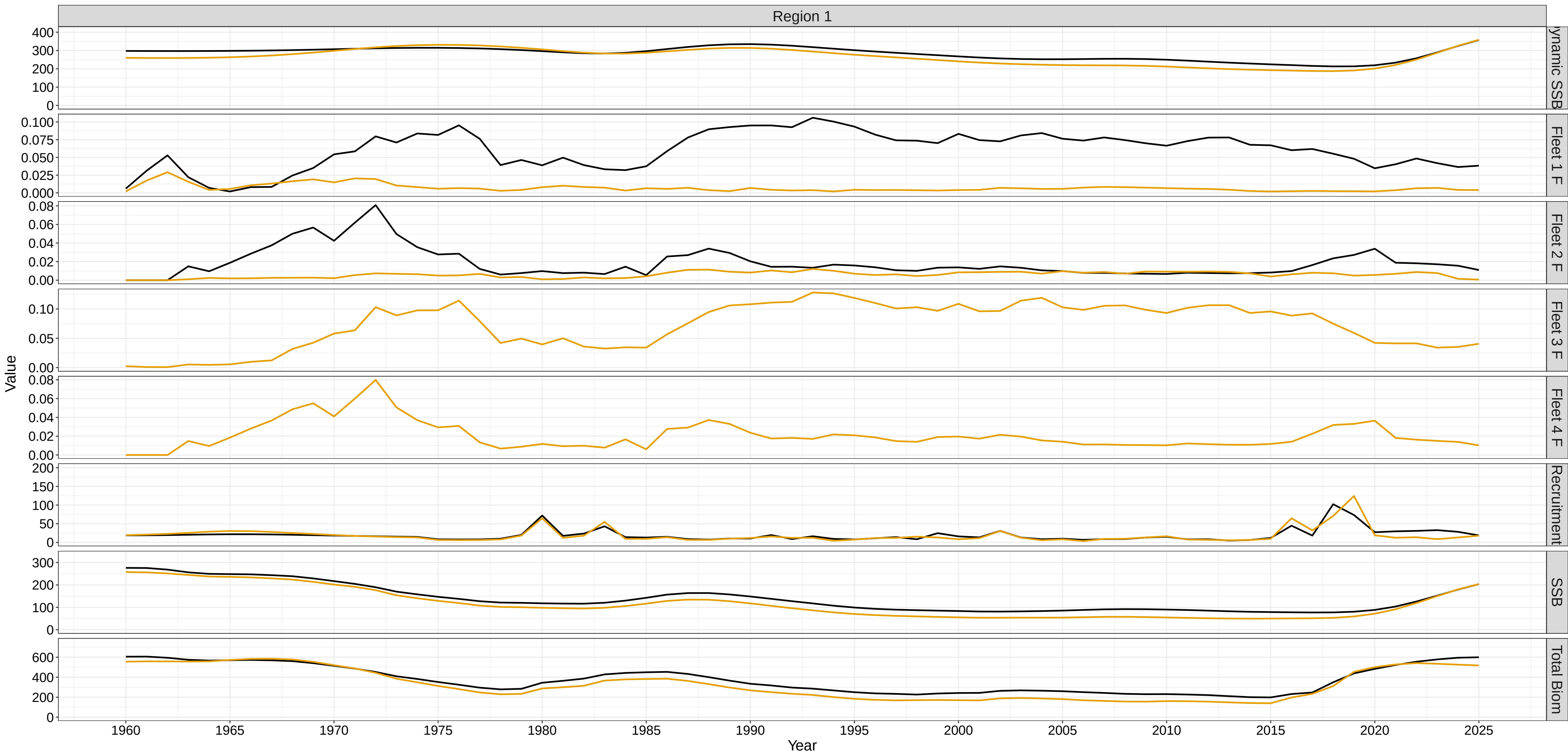
# Survey Selectivity

Model — 26.2c\_FAA\_3Flt      Year 



# Time Series Plots

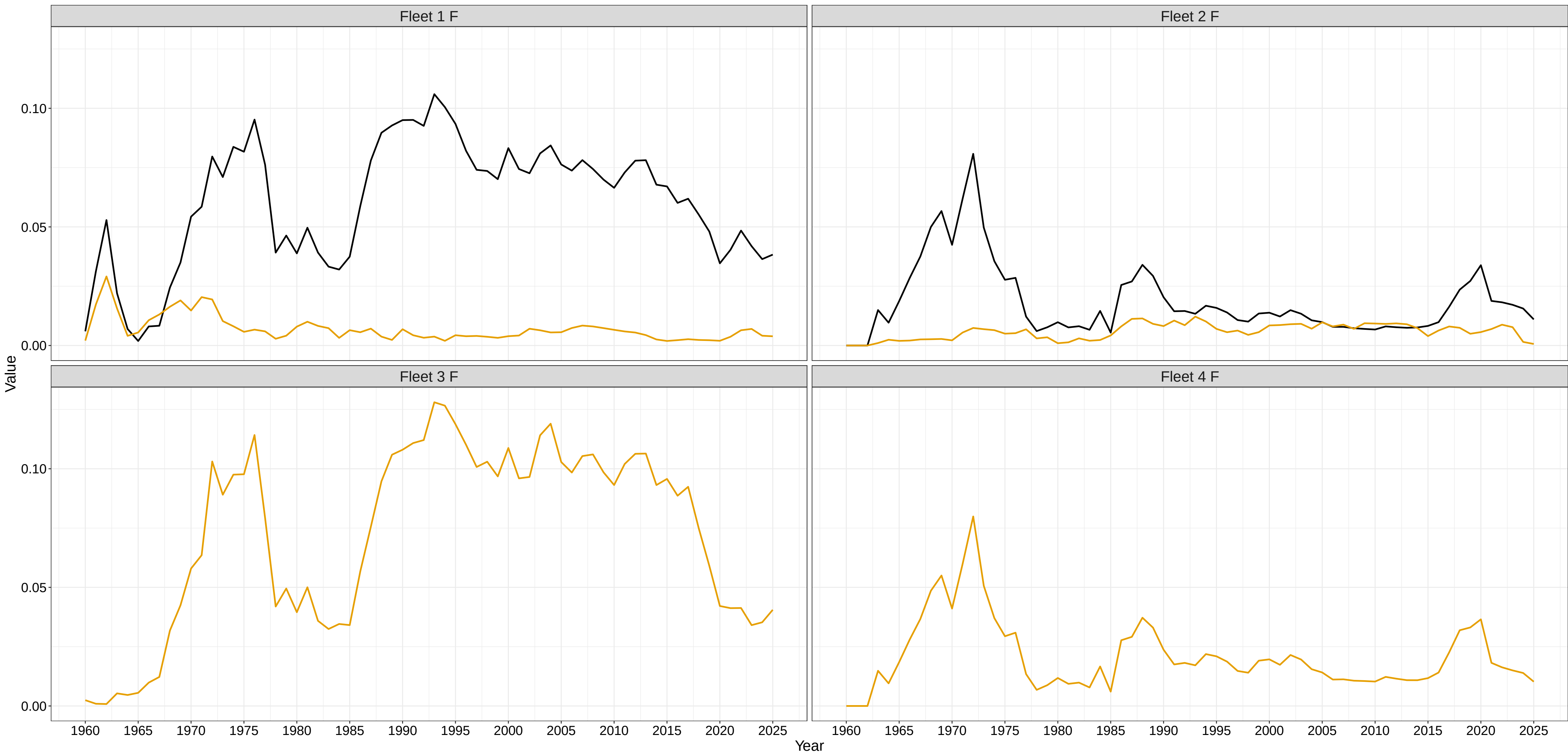
Model — 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS — 26.2c\_FAA\_3Flt





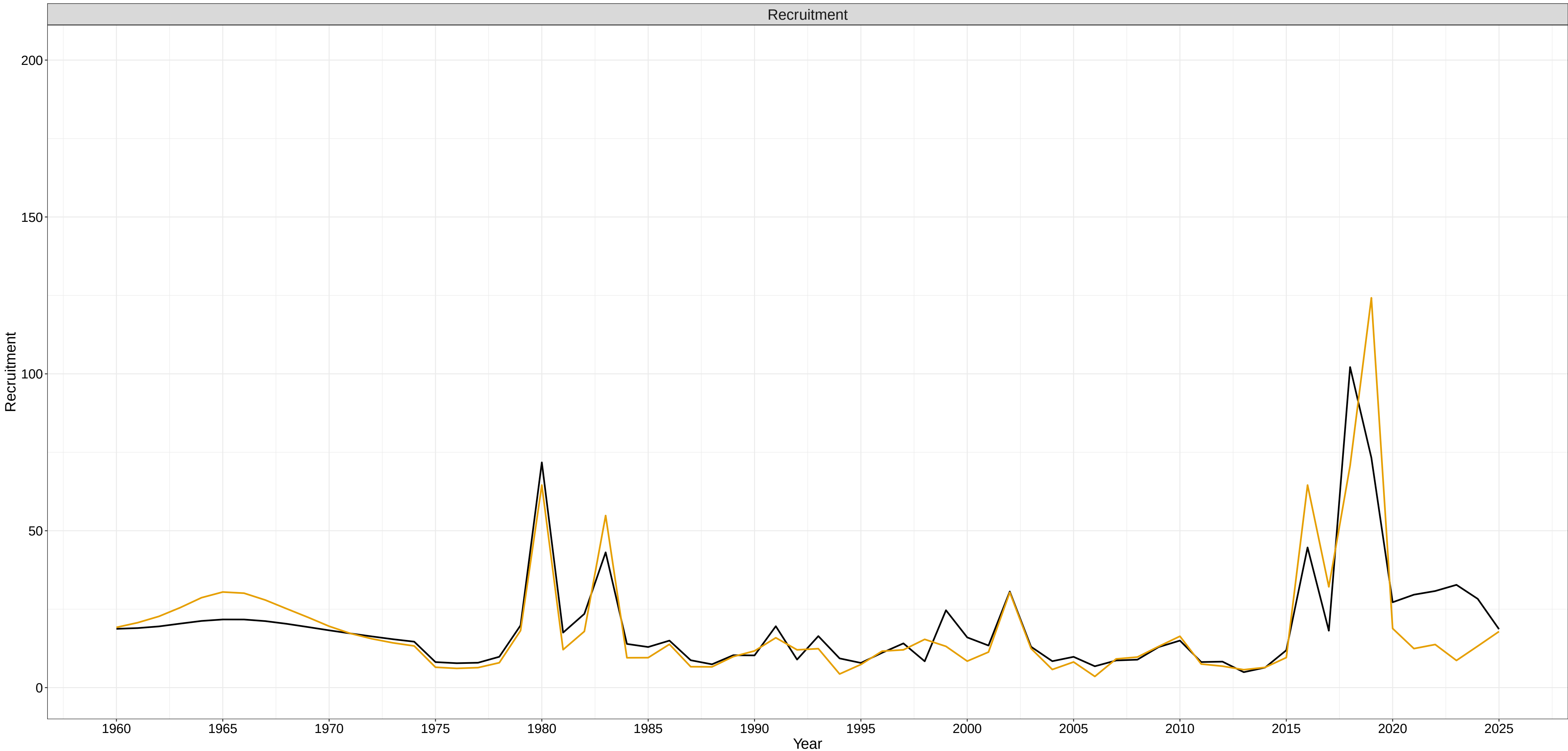
# Fishing Mortality

Model — 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS — 26.2c\_FAA\_3Fit



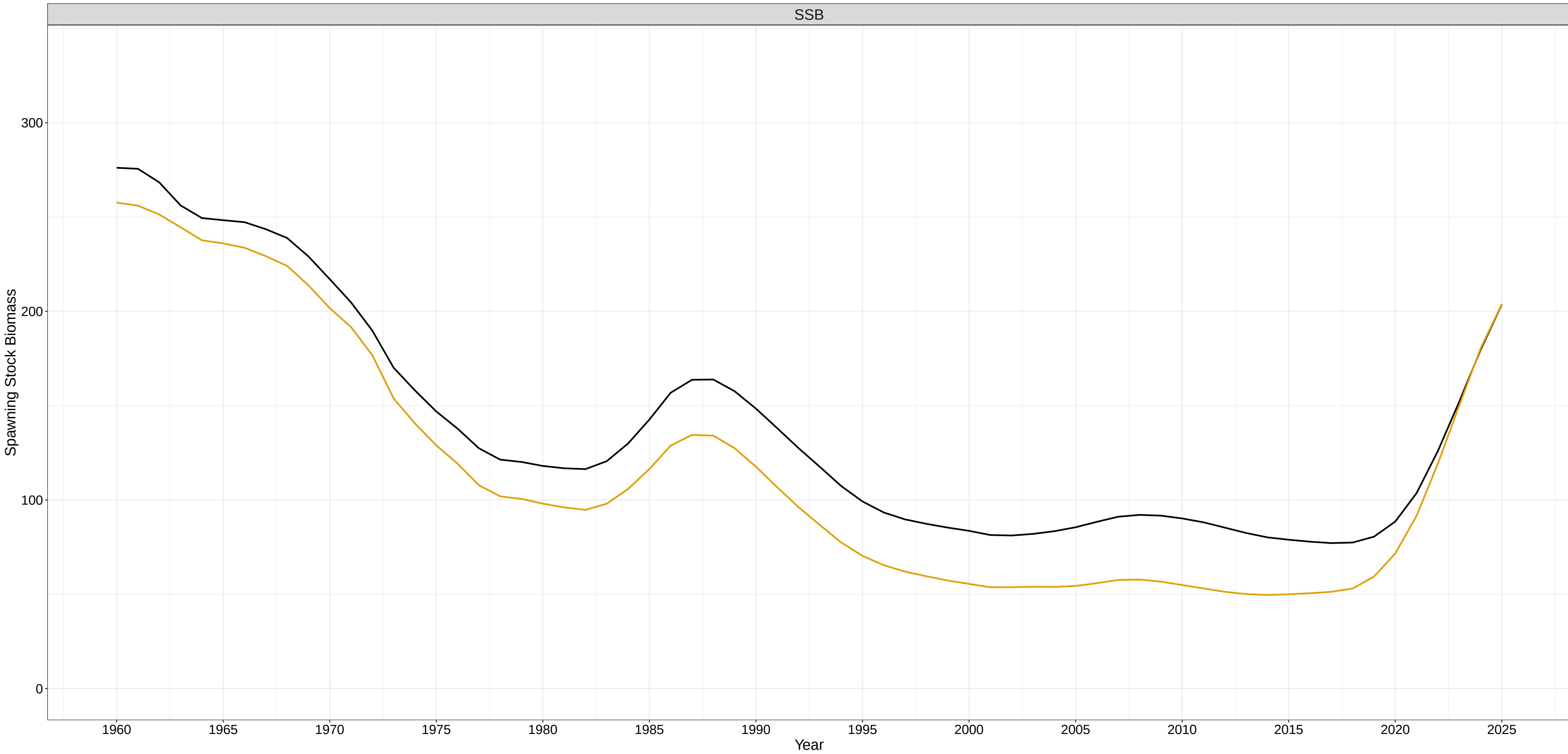
# Recruitment

Model — 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS — 26.2c\_FAA\_3Flt



SSB

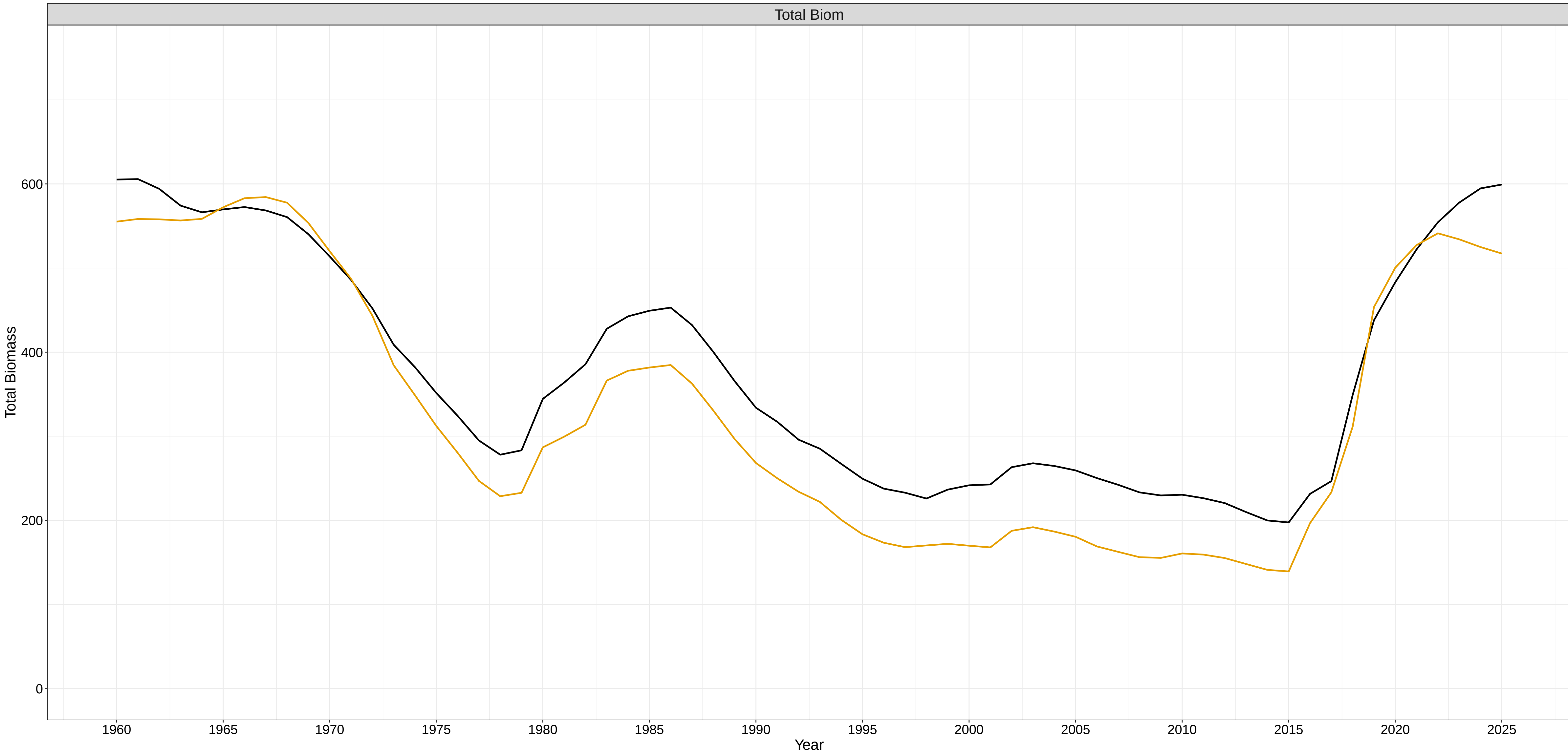
Model — 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS — 26.2c\_FAA\_3Flt





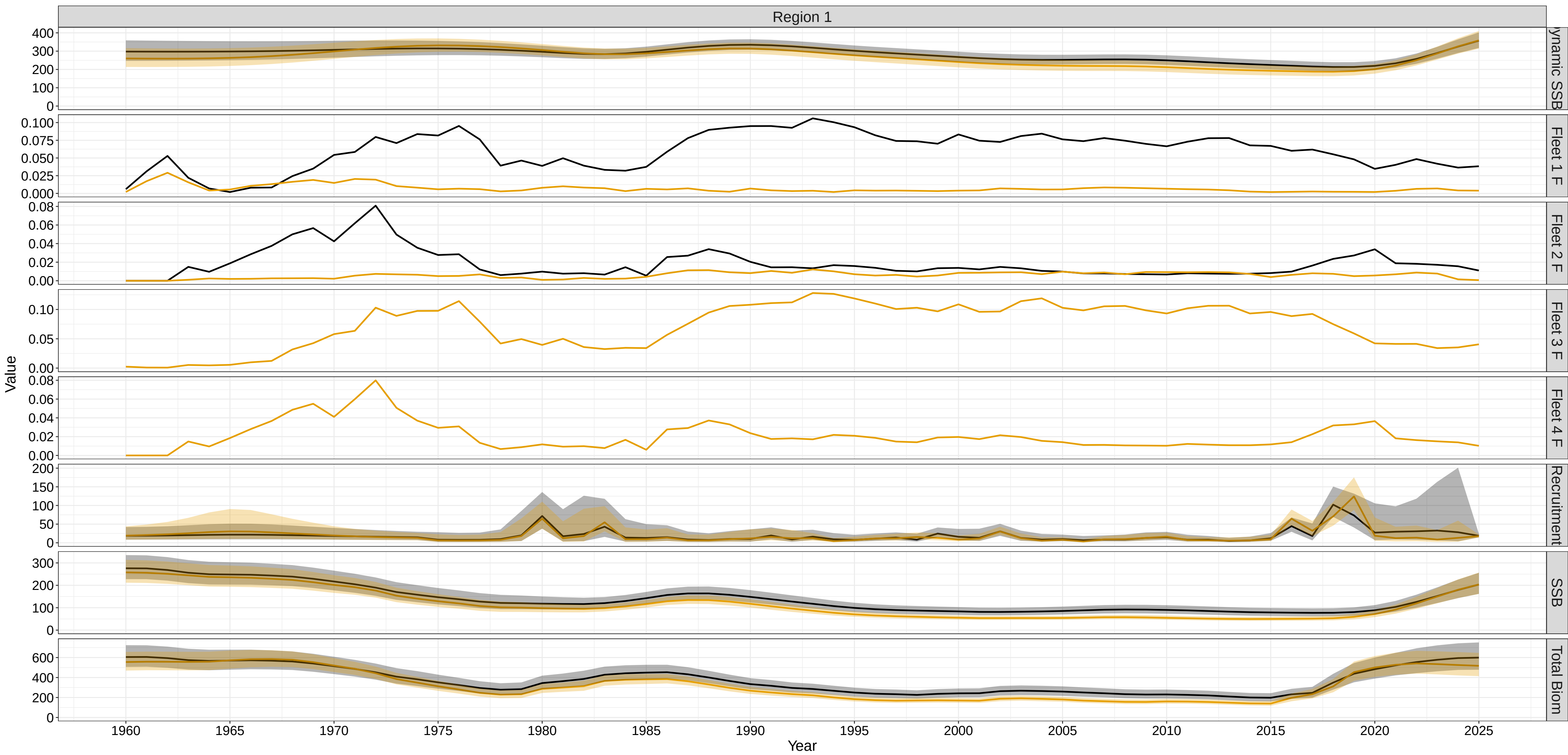
# Biomass

Model — 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS — 26.2c\_FAA\_3Flt



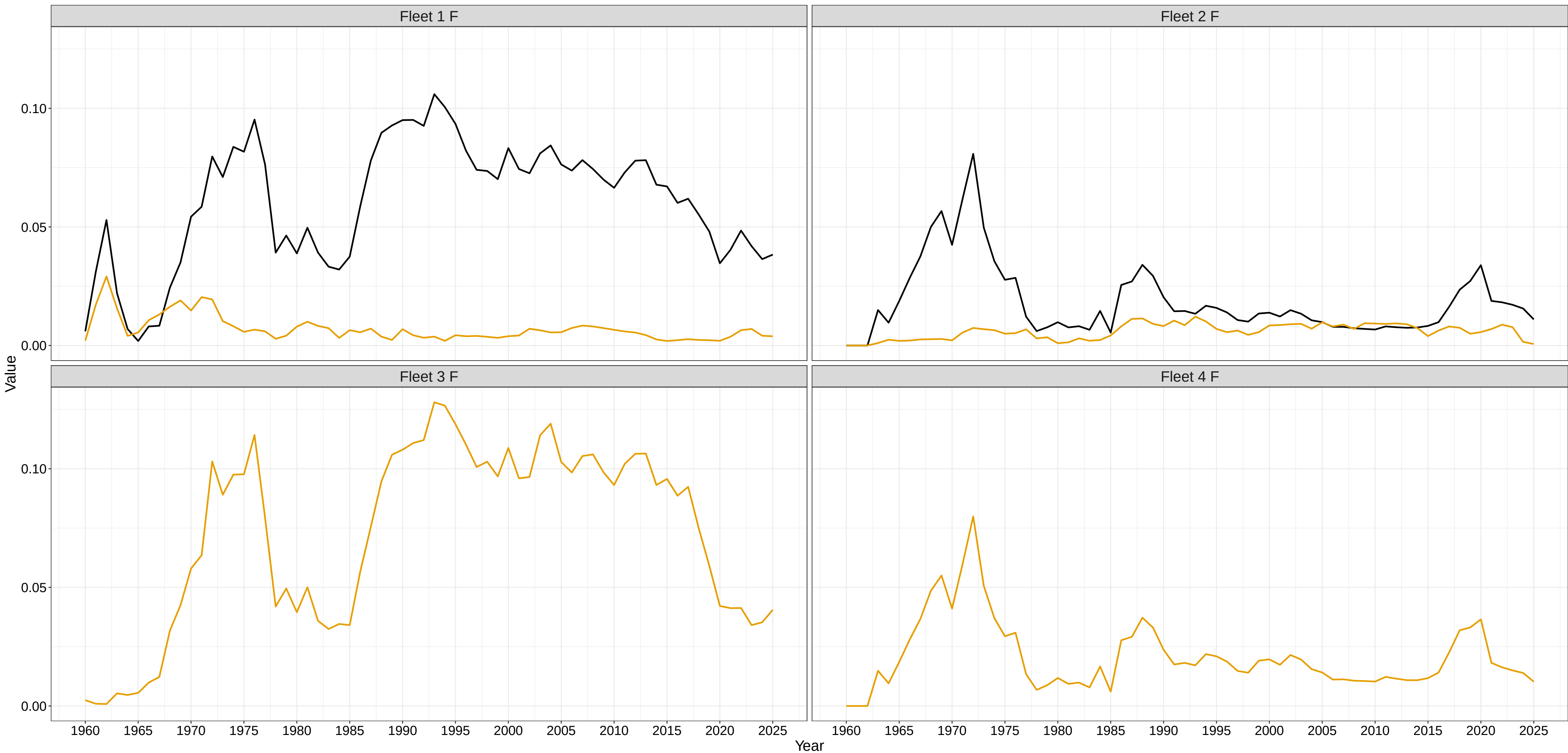
# Time Series Plots

Model 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS 26.2c\_FAA\_3Flt



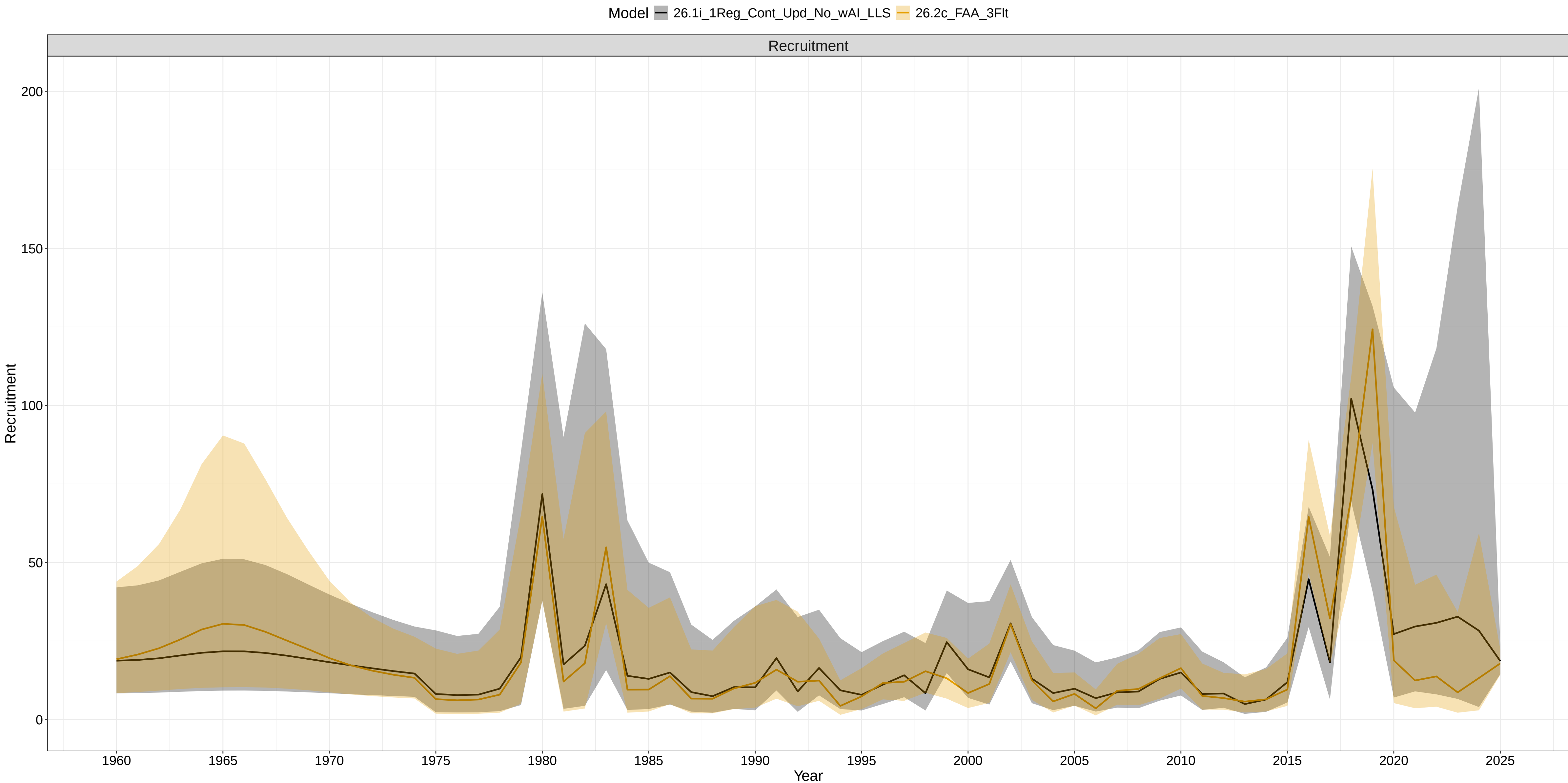
# Fishing Mortality

Model — 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS — 26.2c\_FAA\_3Fit



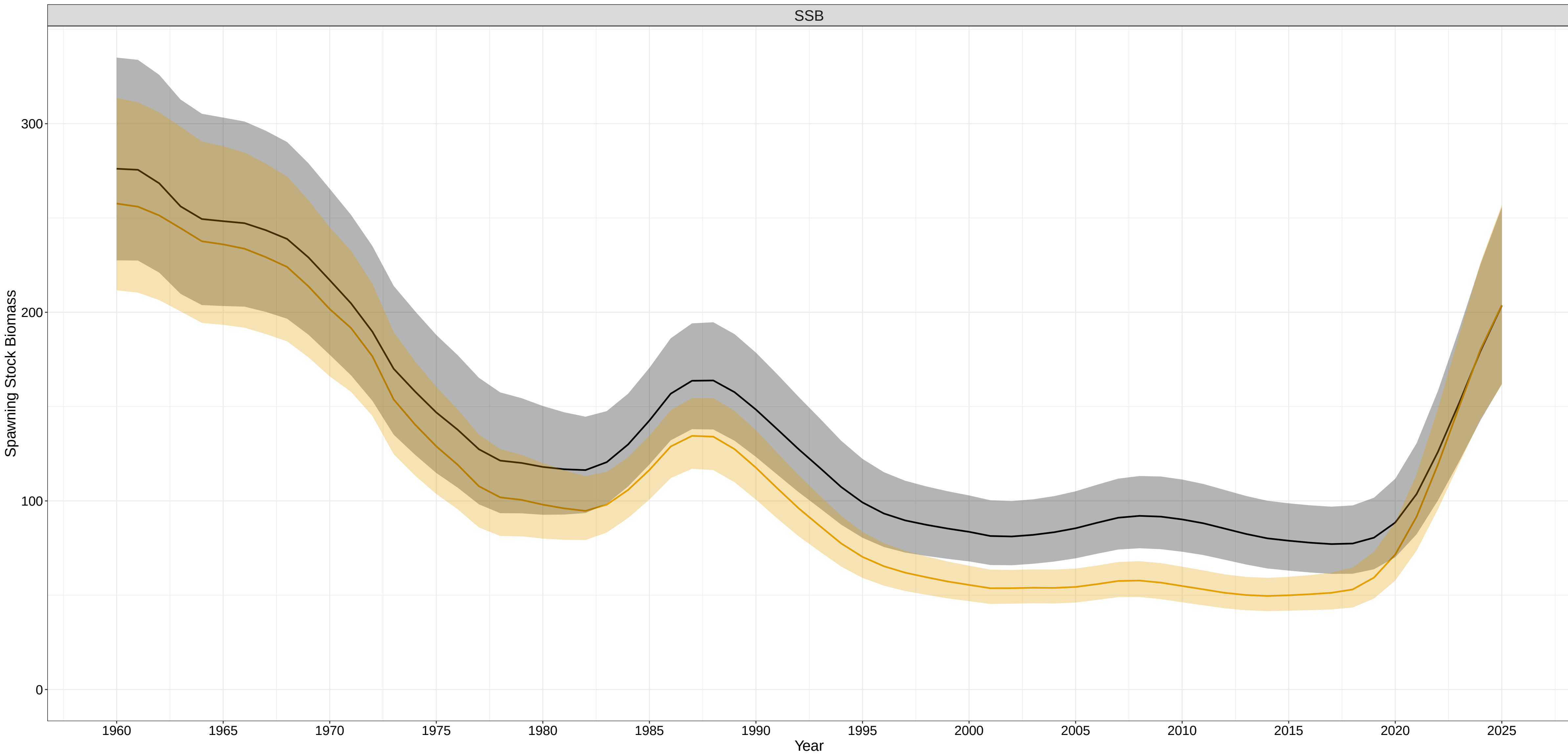


# Recruitment



## SSB

Model 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS 26.2c\_FAA\_3Flt





# Biomass

Model — 26.1i\_1Reg\_Cont\_Upd\_No\_wAI\_LLS — 26.2c\_FAA\_3Flt

